BRUHAT-INCREASING BIJECTION AND BALANCED PERMANENTAL INEQUALITY FOR ALL TNN MATRICES

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Editorial note. Complete original mathematical paper496–5343, all constructive network/bijection proofs, inline diagrams and original image retained. Native skan.bib and dart.bst supplied; wrapper delegates the original bibliography exactly once. First compile needs job-local BSTINPUTS to package-root-r001/items for dart.bst; no bibliography metadata is invented.

Editorial note. The selected main object is one count; family refinements, even/odd cases and its immediate matrix-inequality consequence are supporting material. The other principal object remains complete author context.

Editorial note. Original introduction593 footnote reports an independent Lean-formalized proof encountered during final preparation. This author statement/link is preserved; it is not treated as an attribution that selected author proofs were AI-discovered, and no external formalization or authorship process is investigated/certified. Original Appendix MATLAB-example pointer is also retained without independent replay.

Editorial note. Authors retain responsibility for mathematics. No proof repair, new assumptions, independent mathematical review, counterexample search or AI dependency-chain tracing.

Allowed standard prerequisites (editorial).

- Unit interval orders, totally nonnegative matrices, permanent path-family interpretations, Catalan star networks and Bruhat rank-matrix order. All source-local labeling/network/bijection algorithms, parity subsets and parameter restrictions retained literally. Standard tableau/pairing criterion and Bruhat rank criterion remain original cited facts.. Complete original constructive combinatorial proofs and all native inline diagrams supplied; no reconstruction of prior matrix-inequality proofs or independent Lean/MATLAB verification.

1. Introduction

Given an $n \times n$ matrix $A = (a_{i,j})$ and subsets $I, J \subseteq [n] := \{1, \ldots, n\}$, define the submatrix $A_{I,J} = (a_{i,j})_{i \in I, j \in J}$ of A, and call $\det(A_{I,J})$ a *minor* of A. We call $A \in \mathrm{Mat}_{n \times n}(\mathbb{R})$ *totally nonnegative* if each of its minors is nonnegative.

It is known that the defining inequalities $\det(A_{I,J}) \geq 0$ of totally nonnegative matrices imply many other polynomial inequalities in the matrix entries. Some of these involve *immanants* of A and its submatrices, i.e., linear combinations of the $n!$ monomials

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$\{a_{1,w_1}\cdots a_{n,w_n} \,|\, w \in \mathfrak{S}_n\}$. In particular for any function $f : \mathfrak{S}_n \to \mathbb{R}$, we define the *$f$-immanant of A* [23] to be the polynomial

$$\mathrm{Imm}_f(A) := \sum_{w\in\mathfrak{S}_n} f(w)a_{1,w_1}\cdots a_{n,w_n} \tag{1.1}$$

generalizing the permanent and determinant

$$\mathrm{per}(A) = \sum_{w\in\mathfrak{S}_n} a_{1,w_1}\cdots a_{n,w_n}, \qquad \det(A) = \sum_{w\in\mathfrak{S}_n} \mathrm{sgn}(w)a_{1,w_1}\cdots a_{n,w_n}.$$

In particular, the study of polynomial inequalities satisfied by the entries of Hermitian positive semidefinite matrices led to the following results for each totally nonnegative $n \times n$ matrix A. (See [21, §1] for more details.) Koteljanskii [13] and Marcus [16] showed that we have

$$\det(A) \le a_{1,1}\cdots a_{n,n}, \tag{1.2}$$

$$\mathrm{per}(A) \ge a_{1,1}\cdots a_{n,n}. \tag{1.3}$$

Fan (see [2]) and Lieb [14] showed that for $I \subset [n]$ and its complement $\bar{I} := [n] \smallsetminus I$ we have

$$\det(A) \;\le\; \det(A_{I,I})\det(A_{\bar{I},\bar{I}}), \tag{1.4}$$

$$\mathrm{per}(A) \;\ge\; \mathrm{per}(A_{I,I})\,\mathrm{per}(A_{\bar{I},\bar{I}}). \tag{1.5}$$

Stembridge [24] and Johnson [25, p. 1088] showed that for each $\mathfrak{S}_n$-character χ and the identity element $e \in \mathfrak{S}_n$ we have

$$\det(A) \;\le\; \frac{\mathrm{Imm}_\chi(A)}{\chi(e)}, \tag{1.6}$$

$$\mathrm{per}(A) \;\ge\; \frac{\mathrm{Imm}_\chi(A)}{\chi(e)}. \tag{1.7}$$

Inequalities (1.2) – (1.7) suggest that determinantal and permanental inequalities for totally nonnegative matrices correspond bijectively, with inequality directions reversed. However, this is not in general true. For instance, Koteljanskii [13] showed that for all $I, J \subset [n]$ and for each totally nonnegative matrix A, we have

$$\det(A_{I\cup J,I\cup J})\det(A_{I\cap J,I\cap J}) \;\le\; \det(A_{I,I})\det(A_{J,J}), \tag{1.8}$$

but the second author and Soskin [21, §6] showed the corresponding (reversed) inequality of permanents not to hold in general.

It would therefore be interesting to understand which inequalities change direction when determinants are replaced by permanents. A natural starting point for such an investigation is the second author's result [18, Thm. 3.2] which extends the inequalities (1.4) by providing an *upper bound* for all products $\det(A_{I,I})\det(A_{\bar{I},\bar{I}})$:

$$\det(A_{I,I})\det(A_{\bar{I},\bar{I}}) \;\le\; \det(A_{[n]\cap 2\mathbb{Z},[n]\cap 2\mathbb{Z}})\det(A_{[n]\smallsetminus 2\mathbb{Z},[n]\smallsetminus 2\mathbb{Z}})$$

whenever A is TNN. It is natural to ask whether we can similarly extend the inequalities (1.5) with a *lower bound* for all products $\mathrm{per}(A_{I,I})\,\mathrm{per}(A_{\bar{I},\bar{I}})$:

$$\mathrm{per}(A_{I,I})\,\mathrm{per}(A_{\bar{I},\bar{I}}) \;\ge\; \mathrm{per}(A_{[n]\cap 2\mathbb{Z},[n]\cap 2\mathbb{Z}})\,\mathrm{per}(A_{[n]\smallsetminus 2\mathbb{Z},[n]\smallsetminus 2\mathbb{Z}}). \tag{1.9}$$

In Section 2 we discuss families of combinatorial objects called unit interval orders and descending star networks, and the totally nonnegative matrices associated to each. In Section 3 we combinatorially interpret permanents of these matrices. In Section 4 we define a family of maps on the set of descending star networks and describe the effects that each map has on permanents of the related matrices. This leads to our main result[1] in Section 5 which shows that inequalities of the form (1.9) hold when

(1) the totally nonnegative matrix A is related to a unit interval order,
(2) I is the interval $[h]$ for any $h \in [n]$.

We conjecture the inequalities to hold for all totally nonnegative matrices and $I = [h]$. On the other hand, we show that the inequalities do not hold for general I, even when restricted to matrices of the form (1). We give an extension of the main result in Section 6 and obtain a result of (1.9) for all TNN matrices in Section 7 and 8. We present other open questions in Section 9.

2. Four families of sets counted by Catalan numbers

A simple subset of $n \times n$ totally nonnegative matrices is the set of 0-1 matrices which are weakly increasing in columns, weakly decreasing in rows, and in which all entries on or below the diagonal are 1. Call this set $\mathcal{A}_n$. For example, $\mathcal{A}_4$ consists of the fourteen matrices

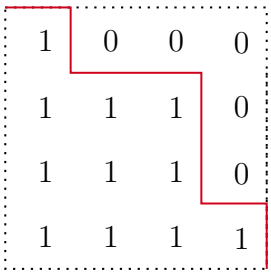
$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix},$$

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \tag{2.1}$$

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

For $A \in \mathcal{A}_n$, the boundary between entries of A equal to 1 and 0 can be viewed as a $2n$-step *Dyck path*, consisting of n horizontal and n vertical steps, none below the diagonal, from the upper left corner of the matrix to the lower right corner. For example, we can draw an 8-step Dyck path on a matrix in $\mathcal{A}_4$.

[1]During the final preparation of this paper, we became aware of an independent proof formalized in Lean. We thank Wenlin Zhang for bringing this work to our attention.



The bijection between $\mathcal{A}_n$ and $2n$-step Dyck paths implies that the cardinality of $\mathcal{A}_n$ is the nth Catalan number, $\frac{1}{n+1}\binom{2n}{n}$. (See [22, Exercise 6.19 (i)].)

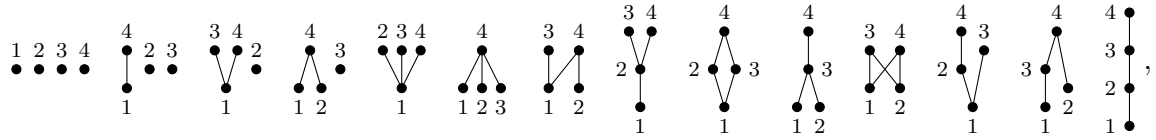
Observation 2.1. *Each matrix in $\mathcal{A}_n$ is totally nonnegative.*

Proof. Fix $A \in \mathcal{A}_n$ and consider a square submatrix B of A. If B has a pair of equal columns or rows, then $\det(B) = 0$. Otherwise, B is unitriangular with $\det(B) = 1$. □

Elements of $\mathcal{A}_n$ correspond bijectively to n-element posets called *unit interval orders*. Call this set $\mathcal{U}_n$. (See [8, p. 33], [22, Exercise 6.19 (ddd)], [27, §8.2].) These are the posets in which no induced four-element subposet is isomorphic to either of the posets

$$\mathbf{3+1} \;=\; \text{  }, \qquad \mathbf{2+2} \;=\; \text{  }.$$

The bijection $A \mapsto P(A)$ is given by declaring $i \leq_{P(A)} j$ if and only if $a_{i,j} = 0$. Each matrix A is said to be the *antiadjacency matrix* of $P(A)$. For example, the fourteen matrices (2.1) are antiadjacency matrices of the unit interval orders

(2.2) 

respectively. Labels, shown here to indicate the correspondence between poset elements and matrix rows and columns, can be computed canonically from an unlabeled unit interval order:

Algorithm 2.2. Given unlabeled unit interval order $P \in \mathcal{U}_n$, do

(1) For each element $y \in P$, compute $\beta(y) := \#\{x \in P \,|\, x \leq_P y\} - \#\{z \in P \,|\, z \geq_P y\}$.
(2) Label the poset elements $1, \ldots, n$ so that we have $\beta(1) \leq \cdots \leq \beta(n)$.

Entries of antiadjacency matrices which correspond to peaks in the Dyck paths mentioned before Observation 2.1 can be interpreted in terms of antichains the corresponding posets.

Proposition 2.3. *Let A be the antiadjacency matrix of unit interval order P, labeled as in Algorithm 2.2. Then an interval $[c, d]$ is a maximal antichain in P if and only if $a_{c,d} = 1$ and this entry is*

(1) the rightmost nonzero entry in row c of A,
(2) the first nonzero entry in column d of A.

Proof. ($\Rightarrow$) We assume that the interval $[c, d]$ is a maximal antichain in P. Then, c and d are incomparable in P, hence $a_{c,d} = 1$. Also, since $[c, d]$ is a maximal, c and $d+j$ are comparable

for all $j > 0$, hence, $[c, d]$ is the rightmost nonzero entry in row c of A. Similarly, $c - i$ and d are comparable for all $i > 0$, hence $a_{c,d}$ is the first nonzero entry in column d of A.

($\Leftarrow$) Next, we assume that $a_{c,d} = 1$ and assume (1) and (2). By the definition of A, c is incomparable to $c + 1, c + 2, \cdots d$. Therefore, $[c, d]$ is an antichain in P. Since $a_{c,d}$ is the rightmost nonzero entry in row c of A, c is comparable to $d+1$, $d+2$, etc. Similarly, since $a_{c,d}$ is the first nonzero entry in column d of A, d is comparable to $c - 1$, $c - 2$, etc. Therefore, $[c, d]$ is a maximal and by the labeling from Algorithm 2.2, it is a maximal antichain in P. □

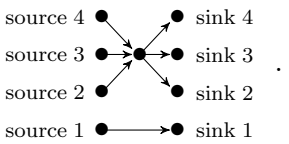
We will show in Section 5 that matrices in $\mathcal{A}_n$ satisfy the inequalities (1.9). To do this, we relate unit interval orders and their antiadjacency matrices to a second family of combinatorical objects counted by Catalan numbers and to a second family of totally nonnegative matrices.

The definition of this second family of combinatorial objects is a bit more intricate than that of unit interval orders. Define a *planar network of order* n to be a directed, planar, acyclic multigraph which can be embedded in a disc so that $2n$ boundary vertices can be labeled clockwise as *source* $1, \ldots,$ *source* n, *sink* $n, \ldots,$ *sink* 1. We will assume all sources to have indegree 0 and outdegree 1, and all sinks to have indegree 1 and outdegree 0. Let $\mathcal{F}_n$ denote the set of such networks.

For each subinterval $[a, b]$ of $[n]$ we define a *simple star network* $F_{[a,b]} \in \mathcal{F}_n$ by

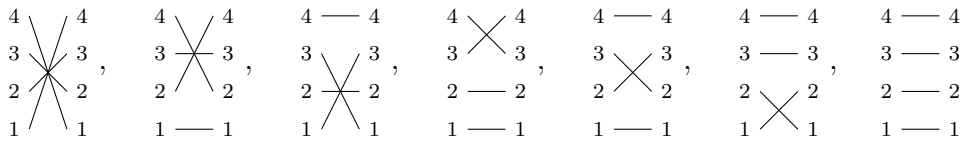
(1) Sources $1, \ldots, n$ lie on a vertical line to the left; sinks $1, \ldots, n$ lie on a vertical line to the right. Both are labeled from bottom to top.
(2) An interior vertex lies between the sources and sinks.
(3) For $i = 1, \ldots, a - 1$ and $i = b + 1, \ldots, n$, a directed edge begins at source i and terminates at sink i.
(4) For $i = a, \ldots, b$, a directed edge begins at source i and terminates at the interior vertex, and another directed edge begins at the interior vertex and terminates at sink i.

For zero- and one-element subintervals we define the trivial network $F_\emptyset = F_{[1,1]} = \cdots = F_{[n,n]}$ to have no interior vertex, and n horizontal edges, each from source i to sink i, for $i = 1, \ldots, n$. For example, the element $F_{[2,4]} \in \mathcal{F}_4$ is



For economy, we will omit edge orientations and vertices from drawings of planar networks. Thus we will draw the seven simple star networks in $\mathcal{F}_4$ as

(2.3)



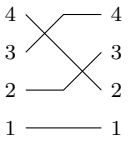
where $F_\emptyset = F_{[1,1]} = F_{[2,2]} = F_{[3,3]} = F_{[4,4]}$. When there is no danger of confusion, we will omit source and sink labels as well.

The set $\mathcal{F}_n$ is generated by simple star networks and two types of concatenation operations. Given networks $E, F \in \mathcal{F}_n$, define the (ordinary) concatenation $E \circ F$ as follows. For $i = 1, \ldots, n$, do

(1) remove sink i of E and source i of F,
(2) merge each edge $(x, \text{sink } i)$ in E with each edge $(\text{source } i, y)$ in F to form a single edge (x, y) in $E \circ F$.

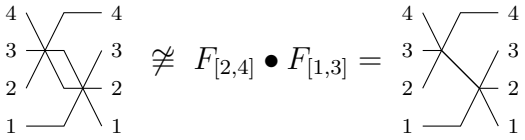
Sometimes in a concatenation $E \circ F$, there may exist vertices x in E, y in F with more than one edge incident upon both. Let $m(x, y)$ be this number. Define the *condensed concatenation* $E \bullet F$ to be the subdigraph of $E \circ F$ obtained by removing, for all such pairs (x, y), all but one of the $m(x, y)$ edges incident upon both. For example, in $\mathcal{F}_4$ we have the isomorphic graphs

$$F_{[3,4]} \circ F_{[2,3]} = \quad \cong \quad F_{[3,4]} \bullet F_{[2,3]}, \tag{2.4}$$



and the nonisomorphic graphs

$$F_{[2,4]} \circ F_{[1,3]} = \quad \ncong \quad F_{[2,4]} \bullet F_{[1,3]} = \quad . \tag{2.5}$$



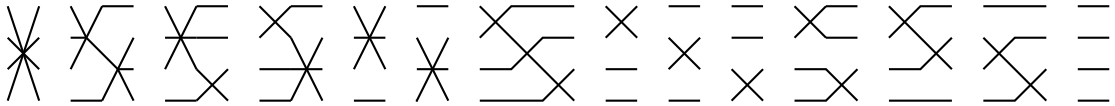
Definition 2.4. Define a *descending star network* to be a condensed concatenation of the form

$$F = F_{[c_1,d_1]} \bullet \cdots \bullet F_{[c_t,d_t]}, \tag{2.6}$$

with intervals $[c_1, d_1], \ldots, [c_t, d_t]$ distinct, pairwise nonnesting, and satisfying $c_1 > \cdots > c_t$ (and $d_1 > \cdots > d_t$).

Let $\mathcal{D}_n$ be the set of all descending star networks of order n. For example, the 14 descending star networks in $\mathcal{D}_4$ are

(2.7)



In the notation of Definition 2.4, these are $F_{[1,4]}$, $F_{[2,4]} \bullet F_{[1,3]}$, $F_{[2,4]} \bullet F_{[1,2]}$, $F_{[3,4]} \bullet F_{[1,3]}$, etc. By [4, Thm. 3.6], [19, Thm. 3.5, Lem 5.3], the number of descending star networks of order n is $\frac{1}{n+1}\binom{2n}{n}$. Explicit bijections $P \mapsto F(P)$ and $F \mapsto P(F)$ between $\mathcal{U}_n$ and $\mathcal{D}_n$ are as follows. (See also [3].)

Algorithm 2.5. Given $P \in \mathcal{U}_n$,

(1) Label P as in Algorithm 2.2.
(2) Let $[c_1, d_1], \ldots, [c_t, d_t]$ be the sequence of maximal antichains in P, ordered so that $c_1 > \cdots > c_t$.
(3) Define $F = F(P) = F_{[c_1,d_1]} \bullet \cdots \bullet F_{[c_t,d_t]}$.

Algorithm 2.6. Given $F \in \mathcal{D}_n$,

(1) For $i = 1, \ldots, n$, define π_i to be the unique path in F from source i to sink i.

(2) Give elements of $P = P(F)$ the labels $1, \ldots, n$.
(3) For all $i, j \in [n]$ with $i < j$ in $\mathbb{Z}$, declare $i <_P j$ if π_i does not intersect π_j in F.

Algorithms 2.5 and 2.6 invert one another. The reader can check that Algorithm 2.5 maps the unit interval orders in (2.2) to the descending star networks in (2.7), respectively, and that Algorithm 2.6 inverts this.

Given a planar network F of order n, define the *path matrix* $B = (b_{i,j})$ of F by

$$b_{i,j} = \text{number of paths in } F \text{ from source } i \text{ to sink } j. \tag{2.8}$$

Let $\mathcal{P}_n$ be the set of $\frac{1}{n+1}\binom{2n}{n}$ path matrices of planar networks in $\mathcal{D}_n$. For example, the set $\mathcal{P}_4$ of path matrices of descending star networks in $\mathcal{D}_4$ (2.7) is

$$\begin{bmatrix} 1&1&1&1\\1&1&1&1\\1&1&1&1\\1&1&1&1 \end{bmatrix}, \begin{bmatrix} 1&1&1&0\\1&1&1&1\\1&1&1&1\\1&1&1&1 \end{bmatrix}, \begin{bmatrix} 1&1&0&0\\1&1&1&1\\1&1&1&1\\1&1&1&1 \end{bmatrix}, \begin{bmatrix} 1&1&1&0\\1&1&1&0\\1&1&1&1\\1&1&1&1 \end{bmatrix}, \begin{bmatrix} 1&0&0&0\\0&1&1&1\\0&1&1&1\\0&1&1&1 \end{bmatrix},$$
$$\begin{bmatrix} 1&1&1&0\\1&1&1&0\\1&1&1&0\\0&0&0&1 \end{bmatrix}, \begin{bmatrix} 1&1&0&0\\1&1&1&0\\1&1&1&1\\1&1&1&1 \end{bmatrix}, \begin{bmatrix} 1&0&0&0\\1&1&0&0\\1&1&1&1\\1&1&1&1 \end{bmatrix}, \begin{bmatrix} 1&0&0&0\\0&1&1&0\\0&1&1&0\\0&0&0&1 \end{bmatrix}, \begin{bmatrix} 1&1&0&0\\1&1&0&0\\0&0&1&0\\0&0&0&1 \end{bmatrix}, \tag{2.9}$$
$$\begin{bmatrix} 1&1&0&0\\1&1&0&0\\0&0&1&1\\0&0&1&1 \end{bmatrix}, \begin{bmatrix} 1&0&0&0\\0&1&1&0\\0&1&1&1\\0&1&1&1 \end{bmatrix}, \begin{bmatrix} 1&1&0&0\\1&1&1&0\\1&1&1&0\\0&0&0&1 \end{bmatrix}, \begin{bmatrix} 1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1 \end{bmatrix}.$$

By Definition 2.4, we may describe the entries of the path matrix of a descending star network in terms of its defining intervals.

Lemma 2.7. *Let descending star network $F = F_{[c_1,d_1]} \bullet \cdots \bullet F_{[c_t,d_t]}$ as in (2.6) have path matrix $B = (b_{i,j})$. Then each entry of B is 0 or 1. In particular, for $i \le j$, we have*

$$b_{i,j} = \begin{cases} 1 & \text{if some interval } [c_m, d_m] \text{ contains both } i \text{ and } j, \\ 0 & \text{otherwise;} \end{cases} \tag{2.10}$$

for $i > j$ we have

$$b_{i,j} = \begin{cases} 1 & \text{if } b_{j,j+1} = b_{j+1,j+2} = \cdots = b_{i-1,i} = 1, \\ 0 & \text{otherwise.} \end{cases} \tag{2.11}$$

Proof. The structure of a descending star network prevents the existence of more than one path from any vertex to any other vertex. Then Equation (2.10) follows immediately from Definition 2.4 and (2.8).

To see the validity of (2.11), observe that by the above definitions we have $b_{i,j} = 1$ with $i > j$ if and only if some indices $m \le m'$ of intervals satisfy $i \in [c_m, d_m]$, $j \in [c_{m'}, d_{m'}]$ and

$$[c_k, d_k] \cap [c_{k+1}, d_{k+1}] \neq \emptyset \text{ for } k = m, \ldots, m' - 1. \tag{2.12}$$

For each such interval $[c_k, d_k]$ there are paths in F from each source $c_k, \ldots, d_k$ to each sink $c_k, \ldots, d_k$. In particular we have $b_{c_k,c_k+1} = b_{c_{k+1},d_{k+2}} = \cdots = b_{d_k-1,d_k} = 1$. Then since

the intervals overlap consecutively, with the first interval containing i and the last interval containing j, we have the right-hand side of (2.11). □

Some, but not all, of the matrices (2.9) appear in (2.1). The precise connection between $\mathcal{A}_n$ and $\mathcal{P}_n$ can be easily described using *block structures* of these matrices and *parabolic subgroups* of $\mathfrak{S}_n$. Let $\alpha = (\alpha_1, \ldots, \alpha_r)$ be an integer composition of n, i.e., a sequence of positive integers satisfying $\alpha_1 + \cdots + \alpha_r = n$, and define the (ordered) set partition $(I_1, \ldots, I_r)$ of $[n]$ into intervals of lengths $\alpha_1, \ldots, \alpha_r$ by

$$I_1 = [1, \alpha_1], \quad I_2 = [\alpha_1 + 1, \alpha_1 + \alpha_2], \quad \ldots, \quad I_r = [n - \alpha_r, n]. \tag{2.13}$$

For any $n \times n$ matrix A, we can write it in terms of *block submatrices* A_{I_i,I_j} by

$$A = (A_{I_i,I_j}) = \begin{bmatrix} A_{I_1,I_1} & A_{I_1,I_2} & \cdots & A_{I_1,I_r} \\ A_{I_2,I_1} & A_{I_2,I_2} & \cdots & A_{I_2,I_r} \\ \vdots & \vdots & & \vdots \\ A_{I_r,I_1} & A_{I_r,I_2} & \cdots & A_{I_r,I_r} \end{bmatrix}. \tag{2.14}$$

If $A_{I_i,I_j} = 0$ whenever $i < j$, we say A is *α-block lower-triangular*; if A and $A^\top$ are both α-block lower-triangular, we say that A is *α-block diagonal*. It is clear that if β is a composition of n with α refining β, then any α-block lower-triangular (diagonal) matrix is also β-block lower-triangular (diagonal). In particular, for any square matrix A there is a finest composition $\alpha = \alpha(A)$ of n for which A is α-block lower-triangular. This composition is determined by the 0-entries of A located just above the diagonal, with the additional property that all entries above and to the right of these are also 0.

Observation 2.8. *Let $A = (a_{i,j})$ be an $n \times n$ 0-1 matrix satisfying $a_{1,j} \leq \cdots \leq a_{j,j}$ for all j, and $a_{i,i} \geq \cdots \geq a_{i,n}$ for all i. Define the set $K = \{k \,|\, a_{k,k+1} = 0\} = \{k_1, \ldots, k_{r-1}\}$ with $k_1 < \cdots < k_{r-1}$, and define $\alpha = \alpha(A)$ as above. Then we have*

$$\alpha(A) = (\alpha_1, \ldots, \alpha_r) = (k_1, k_2 - k_1, \ldots, k_{r-1} - k_{r-2}, n - k_{r-1}). \tag{2.15}$$

Define $\mathfrak{S}_\alpha$ to be the subgroup of $\mathfrak{S}_n$ whose elements have one-line notations of the form $w^{(i)} \circ \cdots \circ w^{(r)}$, where $w^{(j)}$ is a permutation of the letters I_j and $\circ$ denotes concatenation of permutations. Such a subgroup is called *parabolic*. (See, e.g., [1] or [11].) It is known that the determinants of two block lower-triangular matrices are equal if their diagonal blocks are the same. More generally, we have the following.

Proposition 2.9. *Let $A = (A_{I_i,I_j})$ be an $n \times n$ block lower-triangular matrix with intervals $(I_1, \ldots, I_r)$ and composition $\alpha = (\alpha_1, \ldots, \alpha_r)$ as in (2.13), and suppose that for some $w \in \mathfrak{S}_n$ the product $a_{1,w_1} \cdots a_{n,w_n}$ is nonzero. Then we have $w \in \mathfrak{S}_\alpha$.*

Proof. Let j be the smallest index for which the letters in positions I_j of w do not all belong to I_j. Choose any index $k \in I_j$ such that we have $w_k \notin I_j$. We cannot have $w_k \in I_1 \cup \cdots \cup I_{j-1}$, because all of these letters appear in the first $\alpha_1 + \cdots + \alpha_{j-1}$ positions of w. Thus w_k belongs to $I_{j+1} \cup \cdots \cup I_r$. But then the block lower triangularity of A implies that $a_{k,w_k} = 0$, contradicting our assumption that $a_{1,w_1} \cdots a_{n,w_n}$ is nonzero. □

Corollary 2.10. *Let $A = (A_{I_i,I_j})$ be α-block lower-triangular, and let $B = (B_{I_i,I_j})$ be the α-block diagonal matrix satisfying*

$$B_{I_i,I_j} = \begin{cases} A_{I_i,I_j} & \text{if } i = j, \\ 0 & \text{otherwise.} \end{cases} \tag{2.16}$$

Then for every permutation $w = w_1 w_2 \cdots w_n \in \mathfrak{S}_n$, we have

$$a_{1,w_1} \cdots a_{n,w_n} = b_{1,w_1} \cdots b_{n,w_n}. \tag{2.17}$$

Now we can precisely state the relationship between $\mathcal{A}_n$ and $\mathcal{P}_n$. Define the map

$$\begin{aligned} \psi : \mathcal{A}_n &\to \mathcal{P}_n \\ A &\mapsto A_{I_1,I_1} \oplus \cdots \oplus A_{I_r,I_r}, \end{aligned} \tag{2.18}$$

where $(I_1, \ldots, I_r)$ is the set partition associated to the composition $\alpha(A)$ as in (2.13).

Proposition 2.11. *Given unit interval order $P \in \mathcal{U}_n$ with antiadjacency matrix A, the descending star network $F = F(P) \in \mathcal{D}_n$ has path matrix $\psi(A)$.*

Proof. Let P, F, A be as above, $B = (b_{i,j})$ be the path matrix of F, and $B' = (b'_{i,j}) = \psi(A)$.

Consider $i \leq j$. We have $b'_{i,j} = a_{i,j}$. If $b'_{i,j} = a_{i,j} = 1$, then i, j form an antichain in P. Let $[c, d]$ be a maximal antichain containing both. By Algorithm 2.5, the star network F contains $F_{[c,d]}$ as a factor. By Lemma 2.7, there is a path from source i to sink j in F and thus we have $b_{i,j} = 1$. If on the other hand $b'_{i,j} = a_{i,j} = 0$, then we have $i <_P j$. Thus no antichain of P contains both i and j, and by Algorithm 2.5, no interval $[c, d]$ corresponding to a factor $F_{[c,d]}$ of F contains both i and j. By Lemma 2.7, there is no path in F from source i to sink j, and we have $b_{i,j} = 0$.

Now consider $i > j$. Let $\alpha(A) = \alpha(B')$ be the finest partition of n such that A is α-block lower-triangular and B' is α-block diagonal. Define the set partition $(I_1, \ldots, I_t)$ associated to α as in (2.13). If $b'_{i,j} = 1$ then some block $I_m = [k_{m-1}+1, k_m]$ of this set partition contains both i and j. Then

$$I_m = [c_{p_1}, d_{p_1}] \cup [c_{p_1+1}, d_{p_1+1}] \cup \cdots \cup [c_{q_1}, d_{q_1}],$$

for some $p_1, q_1 \in [n]$, where $[c_s, d_s] \cap [c_{s+1}, d_{s+1}] \neq \emptyset$, and $s = p_1, p_1 + 1, \ldots, q_1 - 1$. Then, by Lemma 2.7, there is a path from source i to sink j. Hence, $b_{i,j} = 1$. If on the other hand $b'_{i,j} = 0$, then there are two distinct blocks I_x and I_y with $y < x$ such that $i \in I_x$ and $j \in I_y$. Then,

$$\begin{aligned} I_x &= [c_{p_2}, d_{p_2}] \cup \cdots \cup [c_{q_2}, d_{q_2}] \\ I_y &= [c_{p_3}, d_{p_3}] \cup \cdots \cup [c_{q_3}, d_{q_3}], \end{aligned}$$

for some $p_2, q_2, p_3, q_3 \in [n]$. Since I_x and I_y are distinct blocks, we have $[c_{q_3}, d_{q_3}] \cap [c_{q_2}, d_{q_2}] = \emptyset$. By Lemma 2.7, there is no path from source i to sink j. Hence, $b_{i,j} = 0$.

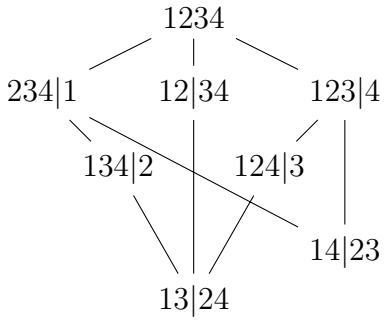
□

3. Permanents, majorization, multinomial coefficients, and path families

Inequalities of the form

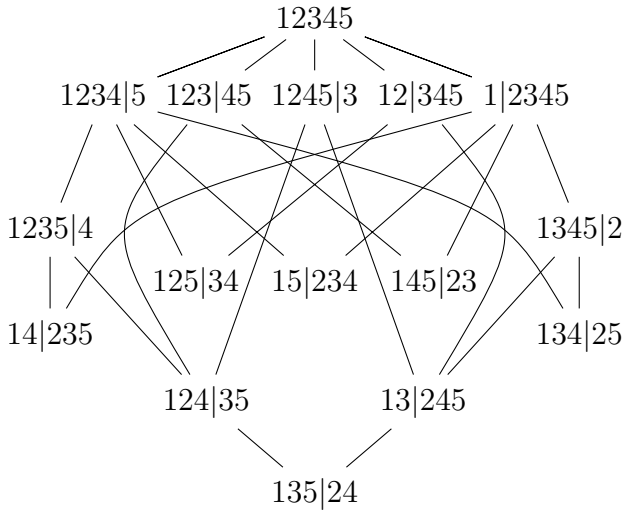
$$\operatorname{per}(A_{L,L}) \operatorname{per}(A_{\overline{L},\overline{L}}) \geq \operatorname{per}(A_{K,K}) \operatorname{per}(A_{\overline{K},\overline{K}}) \tag{3.1}$$

which hold for all $n \times n$ antiadjacency matrices $A \in \mathcal{A}_n$ (Section 2) define a partial order on set partitions $L|\overline{L}$ of $[1,n]$ into at most two blocks: we define $K|\overline{K} \leq L|\overline{L}$ if (3.1) holds for matrices $A \in \mathcal{A}_n$. For example, when $n = 4$, one may evaluate the $2^{4-1} = 8$ products of complementary permanents at the $C_4 = 14$ matrices in $\mathcal{A}_4$ to obtain the Hasse diagram



.

Contrary to our intuition (1.9), we see that the odd-even partition 13|24 is not the unique minimal element of this poset. Similarly, when $n = 5$, one may evaluate the $2^{5-1} = 16$ products of complementary permanents at the $C_5 = 42$ matrices in $\mathcal{A}_5$ to obtain the Hasse diagram



.

Again, the odd-even partition 135|24 is not the unique minimal element of the poset. In both of the above posets, the odd-even set partitions are lower bounds for sets of set partitions consisting of complementary subintervals. For $n = 4$ these are 1234, 123|4, 12|34, 1|234. For $n = 5$ they are 12345, 1234|5, 123|45, 12|345, 1|2345.

The above facts generalize to all n, and we will state the generalization in Theorem 5.1. The proof of this theorem relies upon planar networks and the following facts concerning majorization and products of factorials.

It is well known that the greatest coefficients in the expansion of $(x_1 + \cdots + x_k)^n$ are those of the monomials $x_1^{a_1} \cdots x_k^{a_k}$ having exponents as close to one another as possible.

Lemma 3.1. *Let $(a_1, \ldots, a_k)$ be a weak composition of n and suppose that for some indices h, ℓ we have $a_\ell - a_h \geq 2$. Let $(\hat{a}_1, \ldots, \hat{a}_k)$ be the composition obtained from $(a_1, \ldots, a_k)$ by*

defining

$$\hat{a}_i = \begin{cases} a_i + 1 & \text{if } i = h, \\ a_i - 1 & \text{if } i = \ell, \\ a_i & \text{otherwise.} \end{cases}$$

Then we have

$$\binom{n}{a_1, \ldots, a_k} \leq \binom{n}{\hat{a}_1, \ldots, \hat{a}_k}.$$

Proof. Since $\frac{a_\ell}{a_h+1} \geq 1$, the left-hand side is at most

$$\frac{a_\ell}{a_h+1} \cdot \frac{n!}{a_1!a_2!\cdots a_k!} = \frac{n!}{\hat{a}_1!\cdots\hat{a}_k!} = \binom{n}{\hat{a}_1, \ldots, \hat{a}_k}.$$

□

Corollary 3.2. *Let $(a_1, \ldots, a_{k-1})$, $(b_1, \ldots, b_k)$ be weak compositions of n whose components satisfy $|a_i - a_j| \leq 1$ and $|b_i - b_j| \leq 1$ for all i, j. Then we have*

$$a_1!\cdots a_{k-1}! \geq b_1!\cdots b_k!. \tag{3.2}$$

Proof. If $\max(a_1, \ldots, a_{k-1}) \geq 2$ then we may apply Lemma 3.1 repeatedly to the weak composition $(a_1, \ldots, a_{k-1}, 0)$ to obtain the weak composition $(b_1, \ldots, b_k)$. On the other hand, if $\max(a_1, \ldots, a_{k-1}) \leq 1$, then we also have $\max(b_1, \ldots, b_k) \leq 1$ and (3.2) holds with equality. □

Given a composition $(a_1, \ldots, a_k)$ of n satisfying $|a_i - a_j| \leq 1$ for all i, j, let us say that it has *nearly equal parts*. For example, $(3, 2, 2)$ is a composition of 7 into 3 nearly equal parts. We can see that the composition is unique up to reordering.

Let x and y be partitions of n, we say x is (weakly) majorized by y, written $x \preceq y$, if

$$\sum_{i=1}^{k} x_i \leq \sum_{i=1}^{k} y_i \quad \text{for all } k = 1, \ldots, n.$$

It is well known that the partitions of n under the majorization order form a lattice. We also define $\lambda \prec\cdot\, \mu$ if $\lambda \prec \mu$ and there is no τ such that $\lambda \prec \tau \prec \mu$. The following lemma is due to Hardy, Littlewood and Pólya [17], and is a famous result in majorization theory.

Lemma 3.3. *Let λ, μ be distinct partitions of n. If $\lambda \prec\cdot\, \mu$ then there are indices $i < j$ such that we have $\lambda_i = \mu_i - 1$ and $\lambda_j = \mu_j + 1$, and all other parts of λ and μ are equal.*

The (weak) majorization order gives rise to a factorial inequality for its parts. More generally, we have the following.

Proposition 3.4. *Let partitions $\lambda = (\lambda_1, \ldots, \lambda_s)$, $\mu = (\mu_1, \ldots, \mu_r)$ of n satisfy $\lambda \preceq \mu$. Then we have $\lambda_1!\cdots\lambda_s! \leq \mu_1!\cdots\mu_r!$.*

Proof. By Lemma 3.3, the majorization order contains a saturated chain

$$\lambda = \lambda^{(0)} \prec\cdot\, \lambda^{(1)} \prec\cdot\, \cdots \prec\cdot\, \lambda^{(k)} = \mu$$

in which consecutive partitions differ in exactly two parts. By k applications of Lemma 3.1, we therefore have

$$\binom{n}{\lambda_1, \ldots, \lambda_s} \geq \binom{n}{\mu_1, \ldots, \mu_r},$$

which is equivalent to the claimed inequality. □

We now have an immediate well-known consequence.

Proposition 3.5. *For $a, b \in \mathbb{N}$ with $a + b = m$, we have*

$$a! \cdot b! \geq \left\lfloor \frac{m}{2} \right\rfloor ! \cdot \left\lceil \frac{m}{2} \right\rceil ! \tag{3.3}$$

Proof. Without loss of generality, we assume $a \leq b$. If $a = \lfloor \frac{m}{2} \rfloor$, then there is nothing to prove. If $a \neq \lfloor \frac{m}{2} \rfloor$, then the partition (b, a) majorizes $(\lceil \frac{m}{2} \rceil, \lfloor \frac{m}{2} \rfloor)$. Then (3.3) follows from Proposition 3.4.

□

A function $f : \mathbb{R} \to \mathbb{R}$ is convex if for all $0 \leq t \leq 1$ and distinct $x_1, x_2 \in \mathbb{R}$,

$$f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2). \tag{3.4}$$

In calculus, we know that the "slope" of a convex function is increasing as x increases. In the discrete case, if f is convex, then the difference $d(x) := f(x+1) - f(x)$ increases as x increases.

In addition to the well-known characterizations of the majorization order, we have the following characterization due to Karamata [17, p. 3–13] which involves convex functions.

Lemma 3.6. *(Karamata's Inequality) Given partitions $x = (x_1, \ldots, x_n)$, $y = (y_1, \ldots, y_n)$, we have $y \preceq x$ if and only if for every convex function f, we have*

$$\sum_{i=1}^{n} f(x_i) \geq \sum_{i=1}^{n} f(y_i). \tag{3.5}$$

One can see that the factorial function (equivalently, the Gamma function) is convex. Therefore, Proposition 3.4 is an immediate consequence of Karamata's inequality.

Consider the Karamata sums (3.5), applied to the partition $\rho(n, p)$ of n into p nearly equal parts. For any function $f : \mathbb{R} \to \mathbb{R}$, define

$$G_f(n, p) := \sum_{i=1}^{p} f(\rho(n, p)_i). \tag{3.6}$$

It is straightforward to show that the difference of two such sums for fixed p and consecutive values of n can be expressed as follows:

Lemma 3.7. *Fix $p \leq n$ and any function $f : \mathbb{R} \to \mathbb{R}$. Then, we have*

$$G_f(n+1, p) - G_f(n, p) = f(\lfloor \tfrac{n}{p} \rfloor + 1) - f(\lfloor \tfrac{n}{p} \rfloor). \tag{3.7}$$

Proof. Write $n = pq + r$, then the partition $\rho(n, p)$ has $p - r$ parts of size $\lfloor \frac{n}{p} \rfloor = q$ and r parts of size $\lfloor \frac{n}{p} \rfloor + 1 = q + 1$. Therefore,

$$G_f(n, p) = (p - r)f(q) + rf(q + 1).$$

Note that to move a partition of n into p nearly equal parts to a partition of $n + 1$ into p nearly equal parts, we just pick one part that was of size q and increase it to $q+1$. Therefore,

$$G_f(n + 1, p) - G_f(n, p) = f(q + 1) - f(q).$$

□

For example, the sums $G_f(8,3)$, $G_f(7,3)$ satisfy

$$\begin{gathered}\rho(8,3)=(3,3,2),\quad \rho(7,3)=(3,2,2),\\ G_f(8,3)=f(3)+f(3)+f(2),\quad G_f(7,3)=f(3)+f(2)+f(2)\\ \lfloor\tfrac{7}{3}\rfloor=2,\quad \lfloor\tfrac{7}{3}\rfloor+1=3,\\ G_f(8,3)-G_f(7,3)=f(3)-f(2).\end{gathered}\tag{3.8}$$

If the function f is convex, then we can relate differences (3.7) by the following inequality.

Lemma 3.8. *Fix $p\le n$ and a convex function $f:\mathbb{R}\to\mathbb{R}$. Then, we have*

$$G_f(n+1,p)-G_f(n,p)\ge G_f(n+1,p+1)-G_f(n,p+1).\tag{3.9}$$

Proof. By Lemma 3.7, the above inequality is equivalent to the nonnegativity of

$$\left[f(\lfloor\tfrac{n}{p}\rfloor+1)-f(\lfloor\tfrac{n}{p}\rfloor)\right]-\left[f(\lfloor\tfrac{n}{p+1}\rfloor+1)-f(\lfloor\tfrac{n}{p+1}\rfloor)\right].\tag{3.10}$$

Next, consider the difference $D(\tau):=f(\tau+1)-f(\tau)$. Since f is convex, $D(\tau)$ is nondecreasing (as τ increases). Since $\lfloor\frac{n}{p}\rfloor\ge\lfloor\frac{n}{p+1}\rfloor$, we have

$$D(\lfloor\tfrac{n}{p}\rfloor)-D(\lfloor\tfrac{n}{p+1}\rfloor)\ge 0,$$

which implies (3.10) is nonnegative. □

For example, the differences

$$\begin{aligned}G_f(8,3)-G_f(7,3)&=f(3)-f(2)\\ G_f(8,4)-G_f(7,4)&=f(\lfloor\tfrac{7}{4}\rfloor+1)-f(\lfloor\tfrac{7}{4}\rfloor)=f(2)-f(1)\end{aligned}$$

satisfy

$$G_f(8,3)-G_f(7,3)\ge G_f(8,4)-G_f(7,4)$$

since the convexity of f guarantees that

$$f(3)-f(2)\ge f(2)-f(1).$$

Define $\mathrm{sort}(\lambda,\mu)$ to be the partition obtained from listing the parts of λ and μ (as a multiset) in weakly decreasing order. Somewhat surprisingly, Lemmas 3.6 - 3.8 imply that sorting certain pairs of partitions into nearly equal parts produces two new partitions which are related by majorization.

Proposition 3.9. *Fix $m\ge k>s$ and let κ be the partition of m into s nearly equal parts and μ be the partition of k into s nearly equal parts. Let λ be the partition of m into $s+1$ nearly equal parts and ν be the partition of k into $s+1$ nearly equal parts. Define the partitions*

$$x:=\mathrm{sort}(\kappa,\nu),\qquad y:=\mathrm{sort}(\lambda,\mu)\tag{3.11}$$

of $m+k$ into $2s+1$ parts. Then we have $y\preceq x$.

Proof. Let $\rho(n,p)$ denote the partition of n into p nearly equal parts as before. Then,

$$\begin{gathered}\kappa=\rho(m,s),\ \mu=\rho(k,s)\\ \lambda=\rho(m,s+1),\ \nu=\rho(k,s+1).\end{gathered}$$

We claim that for every convex function f, we have

$$\sum_{i=1}^{2s+1} f(y_i) \leq \sum_{i=1}^{2s+1} f(x_i). \tag{3.12}$$

To see this, define $G_f(n,p)$ as in Lemma 3.7. By Lemma 3.8, we have

$$G_f(n+1,p) - G_f(n+1,p+1) \geq G_f(n,p) - G_f(n,p+1).$$

Since $m \geq k$, we have

$$G_f(m,s) - G_f(m,s+1) \geq G_f(k,s) - G_f(k,s+1),$$

i.e.,

$$G_f(m,s) + G_f(k,s+1) \geq G_f(k,s) + G_f(m,s+1).$$

By the definitions of G_f (3.6) and x, y (3.11), we have

$$G_f(m,s) + G_f(k,s+1) = \sum_{i=1}^{2s+1} f(x_i), \quad G_f(k,s) + G_f(m,s+1) = \sum_{i=1}^{2s+1} f(y_i).$$

Hence, (3.12) is true. By Lemma 3.6, we have proved this proposition.

□

By Proposition 3.4, the majorization of one partition by another implies a factorial inequality for the parts of those two partitions. Applying this to the partitions x, y in (3.11), we obtain the following.

Corollary 3.10. *Fix $s, m, k \in \mathbb{N}$ with $s < k \leq m$. Let $(k_1, \ldots, k_s)$, $(k'_1, \ldots, k'_{s+1})$ be the partitions of k into s nearly equal parts and $s+1$ nearly equal parts respectively. Let $(m_1, \ldots, m_s)$, $(m'_1, \ldots, m'_{s+1})$ the partitions of m into s nearly equal parts and $s+1$ nearly equal parts respectively. Then we have*

$$\frac{m_1! \cdots m_s!}{k_1! \cdots k_s!} \geq \frac{m'_1! \cdots m'_{s+1}!}{k'_1! \cdots k'_{s+1}!}. \tag{3.13}$$

Proof. The result follows immediately from Proposition 3.4 and Proposition 3.9. □

For example, let $m = 11$, $k = 7$, and $s = 3$. Then, we have

$$\begin{array}{lllllll} m_1 = 4 & m_2 = 4 & m_3 = 3 & k'_1 = 2 & k'_1 = 2 & k'_3 = 2 & k'_4 = 1 \\ m'_1 = 3 & m'_2 = 3 & m'_3 = 3 & m'_4 = 2 & k_1 = 3 & k_2 = 2 & k_3 = 2. \end{array}$$

One can verify that

$$\begin{gathered} \operatorname{sort}(m_1, m_2, m_3, k'_1, k'_2, k'_3, k'_4) = (4,4,3,2,2,2,1), \\ \operatorname{sort}(m'_1, m'_2, m'_3, m'_4, k_1, k_2, k_3) = (3,3,3,3,2,2,2), \text{ and} \\ (3,3,3,3,2,2,2) \preceq (4,4,3,2,2,2,1), \\ 144 = \frac{3!4!4!}{2!2!3!} \geq \frac{2!3!3!3!}{1!2!2!2!} = 54. \end{gathered}$$

The next proposition is a special case of Corollary 3.10.

Proposition 3.11. *For $m, k \in \mathbb{N}$ with $m \geq k$, we have*

$$\frac{\lfloor \frac{m}{2} \rfloor ! \lceil \frac{m}{2} \rceil !}{\lfloor \frac{k}{2} \rfloor ! \lceil \frac{k}{2} \rceil !} \leq \frac{m!}{k!}.$$

Proof. This is a special case of Corollary 3.10 as $s = 1$. □

We next prove a stronger version of the previous proposition. One can see that this is no longer a special case of Corollary 3.10.

Proposition 3.12. *For $m, k \in \mathbb{N}$ with $m \geq k$, we have*

$$\frac{\lfloor \frac{m}{2} \rfloor ! \lceil \frac{m}{2} \rceil !}{\lfloor \frac{k}{2} \rfloor ! \lceil \frac{k}{2} \rceil !} \leq \frac{(m-1)!}{(k-1)!}. \tag{3.14}$$

Proof. If $m = k$, (3.14) is trivially true. Suppose therefore that $k < m$. Multiplying both sides of (3.14) by $(m+k-1)!$, we obtain the equivalent inequality

$$\binom{m+k-1}{m-1, \lceil \frac{k}{2} \rceil, \lfloor \frac{k}{2} \rfloor} \leq \binom{m+k-1}{k-1, \lceil \frac{m}{2} \rceil, \lfloor \frac{m}{2} \rfloor}. \tag{3.15}$$

By Lemma 3.1, we have

$$\begin{aligned} \binom{m+k-1}{m-1, \lceil \frac{k}{2} \rceil, \lfloor \frac{k}{2} \rfloor} &\leq \binom{m+k-1}{m-2, \lceil \frac{k+1}{2} \rceil, \lfloor \frac{k+1}{2} \rfloor} \leq \cdots \\ &\cdots \leq \binom{m+k-1}{m-1-(m-k), \lceil \frac{k+(m-k)}{2} \rceil, \lfloor \frac{k+(m-k)}{2} \rfloor}, \end{aligned}$$

since for $i = 0, 1, \ldots, m-k$, we have $\lfloor \frac{k+i}{2} \rfloor, \lceil \frac{k+i}{2} \rceil \leq k+i \leq m+i-1$ and

$$\left(\left\lfloor \frac{k+1}{2} \right\rfloor, \left\lceil \frac{k+i}{2} \right\rceil \right) = \begin{cases} (\lfloor \frac{k+i}{2} \rfloor, \lceil \frac{k+i-1}{2} \rceil + 1) \text{ when } k+i \text{ is even,} \\ (\lceil \frac{k+i-1}{2} \rceil, \lfloor \frac{k+i-1}{2} \rfloor + 1) \text{ when } k+i \text{ is odd.} \end{cases}$$

□

For example, let $m = 11$ and $k = 7$. Then we have

$$m-1 = 10, \quad k-1 = 6,$$
$$\lfloor \tfrac{m}{2} \rfloor = 5, \quad \lceil \tfrac{m}{2} \rceil = 6, \quad \lfloor \tfrac{k}{2} \rfloor = 3, \quad \lceil \tfrac{k}{2} \rceil = 4.$$

One can verify that

$$\frac{(m-1)!}{(k-1)!} = \frac{10!}{6!} = 5040 \geq \frac{\lfloor \frac{m}{2} \rfloor ! \lceil \frac{m}{2} \rceil !}{\lfloor \frac{k}{2} \rfloor ! \lceil \frac{k}{2} \rceil !} = \frac{5!6!}{3!4!} = 600.$$

It would be interesting to describe the analogous poset defined by inequalities (3.1) holding for *all* totally nonnegative matrices A (not just those in $\mathcal{A}_n$). This problem appears to be more difficult than its determinantal analog solved in [7]. (See also [18].) We therefore restrict our attention to the set $\mathcal{A}_n$. Indeed other total nonnegativity results are more easily proved for $A \in \mathcal{A}_n$, then for A totally nonnegative. For instance, compare the proofs [9, Thm. 2] and [24, Cor.3.2] of the inequalities $\mathrm{Imm}_\chi(A) \geq 0$ which hold for $A \in \mathcal{A}_n$, and A totally nonnegative, respectively. To study this poset induced by (3.1), we look more closely at the planar networks defined in Section 2 and their connection to permanents.

Let $\pi = (\pi_1, \ldots, \pi_n)$ be a sequence of source-to-sink paths in a planar network $F \in \mathcal{F}_n$. We call π a *path family* if there exists a permutation $w = w_1 \cdots w_n \in \mathfrak{S}_n$ such that π_i is a path from source i to sink w_i. In this case, we say that π has *type* w. Further, we say that the path family *covers* F if it contains every edge exactly once.

If $A \in \mathrm{Mat}_{n\times n}$ is the path matrix of $F \in \mathcal{F}_n$, then some immanants (1.1) of A can be interpreted in terms of path families in R. For instance the simple immanant $a_{1,w_1} \cdots a_{n,w_n}$ equals the number of path families of type w in F.

Let F be a planar network of order n and let I, J be subsets of $[n]$ with $|I| = |J|$. Let $\Pi_{I,J}(F)$ denote the set of path families from sources indexed by I to sinks indexed by J. For economy, let $\Pi(F) = \Pi_{[n],[n]}(F)$.

The following interpretation of the determinant is due to Karlin and McGregor [12] and usually attributed to Lindstrom [15].

Lemma 3.13. *We have that* $\det(A)$ *equals the number of path families* π *in* F *in which* $\pi_1, \ldots, \pi_n$ *are pairwise nonintersecting.*

Corollary 3.14. *For* $I, J \subseteq [n]$ *with* $|I| = |J| = k$*, we have*

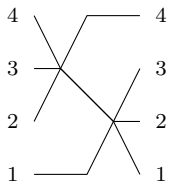
$$\det(A_{I,J}) = \#\{\pi = (\pi_1, \ldots, \pi_k) \in \Pi_{I,J}(F) \,|\, \pi_1, \ldots, \pi_k \text{ pairwise nonintersecting } \}.$$

We have the following combinatorial interpretation of the permanent using path families.

Observation 3.15. *Let* A *be the path matrix of planar network* $F \in \mathcal{F}_n$*. Then* $\mathrm{per}(A)$ *is the number of path families in* F*.*

Corollary 3.16. *For* $I, J \subseteq [n]$ *with* $|I| = |J| = k$*, we have* $\mathrm{per}(A_{I,J}) = |\Pi_{I,J}(F)|$.

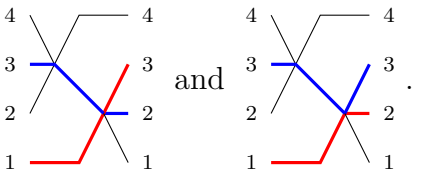
For example, the planar network F



has path matrix

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

We see that $\mathrm{per}(A_{\{1,3\},\{2,3\}}) = 2$. (Note that we define $\mathrm{per}(A_{\emptyset,\emptyset}) = 1$.) The 2 path families in $\Pi_{\{1,3\},\{2,3\}}(F)$ are



There is a simple formula which counts path families covering a descending star network.

Lemma 3.17. *Given descending star network* $F = F_{[c_1,d_1]} \bullet \cdots \bullet F_{[c_r,d_r]}$*, define numbers* $m_1, \ldots, m_r$*,* $k_1, \ldots, k_{r-1}$ *by*

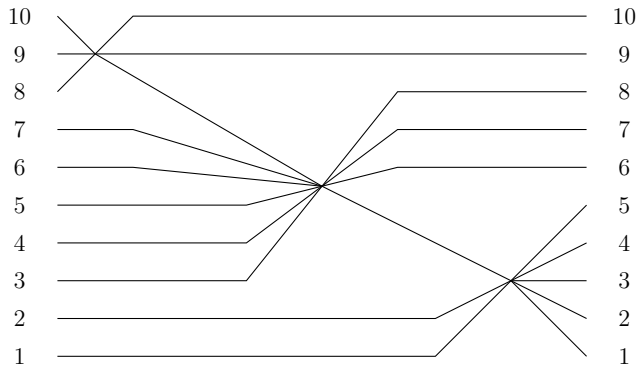
$$\begin{aligned} m_i &= d_i - c_i + 1, \\ k_i &= \max\{d_{i+1} - c_i + 1, 0\}. \end{aligned} \tag{3.16}$$

Then we have

$$|\Pi_{[n],[n]}(F)| = \frac{m_1! \cdots m_{r-1}!}{k_1! \cdots k_{r-1}!} \cdot m_r!. \tag{3.17}$$

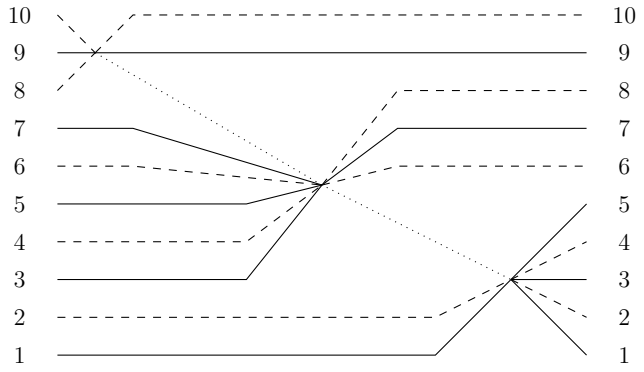
Proof. If F is a simple star network $F = F_{[c_1,d_1]}$. Then $|\Pi_{[n],[n]}(F)|$ is clearly $(d_1 - c_1 + 1)!$. Now suppose (3.17) holds for descending star networks with $1, \ldots, r-1$ factors and consider F with r factors. Let $F' = F_{[c_1,d_1]} \bullet \cdots \bullet F_{[c_{r-1},d_{r-1}]}$, $F = F' \bullet F_{[c_r,d_r]}$ and consider $\Pi(F' \circ F_{[c_r,d_r]})$. Clearly $|\Pi(F' \circ F_{[c_r,d_r]})| = |\Pi(F')| \cdot |\Pi(F_{[c_r,d_r]})| = \frac{m_1! \cdots m_{r-1}!}{k_1! \cdots k_{r-2}!} \cdot m_r!$. Now recall that $F' \bullet F_{[c_r,d_r]}$ is obtained from $F' \circ F_{[c_r,d_r]}$ by removing all but one of the k_{r-1} edges from central vertices z_{r-1} to z_r. Call this edge e. This gives a surjective map $\pi \mapsto \widehat{\pi}$ from $\Pi(F \circ F_{[c_r,d_r]}$ to $\Pi(F \bullet F_{[c_r,d_r]})$ by $\widehat{\pi} = \pi_i$ with its z_{r-1} to z_r edge replaced by e. It is clear that the preimage of any path family $\widehat{\pi} \in \Pi(F' \bullet F_{[c_r,d_r]})$ has cardinality $k_{r-1}!$. Thus we have the claimed formula for $|\Pi(F' \bullet F_{[c_r,d_r]})|$. □

For example, when $F = F_{[8,10]} \bullet F_{[3,8]} \bullet F_{[1,5]}$, paths in $\Pi_{[10],[10]}(F)$ are the paths covering the edges in



By (3.17), we see that $|\Pi_{[10],[10]}(F) = \frac{3!6!5!}{1!3!}$.

Now consider the variation of (3.17) which counts paths in $\Pi_{[n],[n]}(F)$ having sources and sinks with only odd (only even) indices. For example, when $F = F_{[8,10]} \bullet F_{[3,8]} \bullet F_{[1,5]}$, these are the paths covering the solid and dotted (dashed and dotted) edges in



We see that $|\Pi_{\{1,3,\ldots,9\},\{1,3,\ldots,9\}}(F)| = \frac{1!3!3!}{0!2!}$ and $|\Pi_{\{2,4,\ldots,10\},\{2,4,\ldots,10\}}(F)| = \frac{2!3!2!}{1!1!}$.
More generally, we have the following.

Lemma 3.18. *Let $F = F_{[c_1,d_1]} \bullet \cdots \bullet F_{[c_{r+1},d_{r+1}]}$ be a descending star network with m_i, k_i defined as* (3.16)*. Define*

$$I := 2\mathbb{Z} \cap [n], \qquad \bar{I} := [n] \smallsetminus I.$$

Then we have

$$|\Pi_{I,I}(F)| \cdot |\Pi_{\bar{I},\bar{I}}(F)| = \frac{\lfloor \frac{m_1}{2} \rfloor! \lceil \frac{m_1}{2} \rceil! \cdots \lfloor \frac{m_{r+1}}{2} \rfloor! \lceil \frac{m_{r+1}}{2} \rceil!}{\lfloor \frac{k_1}{2} \rfloor! \lceil \frac{k_1}{2} \rceil! \cdots \lfloor \frac{k_r}{2} \rfloor! \lceil \frac{k_r}{2} \rceil!}. \tag{3.18}$$

Proof. Observe that the product on the left-hand side of (3.18) equals the cardinality of path families $\pi = (\pi_1, \ldots, \pi_n)$ covering F such that for all i, the number $\mathrm{snk}(\pi_i)$ has the same parity as i. Call this set $\Pi_{\mathrm{par}}(F)$ so that $|\Pi_{I,I}(F)| \cdot |\Pi_{\bar{I},\bar{I}}(F)| = |\Pi_{\mathrm{par}}(F)|$.

Observe that when F is the simple star $F_{[c_1,d_1]}$, we have $|\Pi_{\mathrm{par}}(F)| = \lfloor \frac{m_1}{2} \rfloor! \lceil \frac{m_1}{2} \rceil!$ and (3.18) is true.

Now suppose that the equation is true for descending star networks which are concatenations of at most r stars, and consider a concatenation

$$F = F_{[c_1,d_1]} \bullet \cdots \bullet F_{[c_{r+1},d_{r+1}]} \tag{3.19}$$

of $r+1$ stars. Define $F' = F_{[c_1,d_1]} \bullet \cdots \bullet F_{[c_r,d_r]}$ so that we have $F = F' \bullet F_{[c_{r+1},d_{r+1}]}$.

Clearly each element of $\Pi_{\mathrm{par}}(F)$ can be obtained by

(1) concatenating an element of $\Pi_{\mathrm{par}}(F')$ with an element of $\Pi_{\mathrm{par}}(F_{[c_{r+1},d_{r+1}]})$ in $F' \circ F_{[c_{r+1},d_{r+1}]}$,
(2) collapsing the k_r edges from the central vertex of $F_{[c_r,d_r]}$ to $F_{[c_{r+1},d_{r+1}]}$ into a single edge (z_r, z_{r+1}) of F which is covered by the k_r corresponding path segments.

Now observe that for any permutation $w \in \mathfrak{S}_n$, the number of path families of type w constructed in step (1) above equals $\lfloor \frac{k_r}{2} \rfloor! \lceil \frac{k_r}{2} \rceil!$ times the number of path families (0 or 1) of type w in $\Pi_{\mathrm{par}}(F)$ after the collapsing done in step (2). It follows that we have

$$\begin{aligned}
|\Pi_{\mathrm{par}}(F)| &= \frac{1}{\lfloor \frac{k_r}{2} \rfloor! \lceil \frac{k_r}{2} \rceil!} |\Pi_{\mathrm{par}}(F' \circ F_{[c_{r+1},d_{r+1}]})| \\
&= \frac{1}{\lfloor \frac{k_r}{2} \rfloor! \lceil \frac{k_r}{2} \rceil!} |\Pi_{\mathrm{par}(F')}| |\Pi_{\mathrm{par}}(F_{[c_{r+1},d_{r+1}]})| \\
&= \frac{1}{\lfloor \frac{k_r}{2} \rfloor! \lceil \frac{k_r}{2} \rceil!} \frac{\lfloor \frac{m_1}{2} \rfloor! \lceil \frac{m_1}{2} \rceil! \cdots \lfloor \frac{m_r}{2} \rfloor! \lceil \frac{m_r}{2} \rceil!}{\lfloor \frac{k_1}{2} \rfloor! \lceil \frac{k_1}{2} \rceil! \cdots \lfloor \frac{k_{r-1}}{2} \rfloor! \lceil \frac{k_{r-1}}{2} \rceil!} \cdot \left\lfloor \frac{m_{r+1}}{2} \right\rfloor! \left\lceil \frac{m_{r+1}}{2} \right\rceil!
\end{aligned}$$

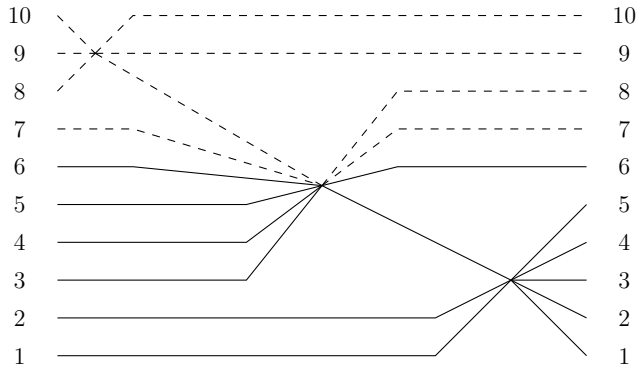
as desired.

□

We keep using the definitions of m_i and k_i as in Lemma 3.18. Let $h \in [1, n]$, then define sets

$$\begin{aligned}
J &= [h], & I &= [n] \cap 2\mathbb{Z}, \\
\overline{J} &= [h+1, n], & \overline{I} &= [n] \smallsetminus 2\mathbb{Z}.
\end{aligned} \tag{3.20}$$

Now consider the variation of (3.17) which counts paths in $\Pi_{[h],[h]}(F)$ when F is a descending star network of order n and $h < n$. For example, when $F = F_{[8,10]} \bullet F_{[3,8]} \bullet F_{[1,5]}$ and $h = 6$, these are the paths covering the solid edges in



We see that $|\Pi_{[6,6]}(F)| = \frac{(6-3+1)!}{3!} \cdot 5!$.

More generally, we have the following.

Lemma 3.19. *Fix $h \in [n]$ and let $F = F_{[c_1,d_1]} \bullet \cdots \bullet F_{[c_r,d_r]}$ be a descending star network such that $F_{[c_p,d_p]}$ is the only star with $[c_p, d_p] \cap [h, h+1] \neq \emptyset$. Then*

$$|\Pi_{[h],[h]}(F)| \cdot |\Pi_{[h+1,n],[h+1,n]}(F)| = \frac{(h - c_p + 1)!(d_p - h)! m_1! \cdots m_{p-1}! m_{p+1}! \cdots m_r!}{k_1! \cdots k_{p-1}! k_{p+1}! \cdots k_{r-1}!}.$$

Proof. Observe that the number $|\Pi_{[h],[h]}(F)|$ is $\frac{(h-c_p+1)!}{k_p!} \frac{m_{p+1}!}{k_{p+1}!} \cdots \frac{m_{r-1}!}{k_{r-1}!} \cdot m_r!$. Similarly, the number $|\Pi_{[h+1,n],[h+1,n]}(F)|$ is $\frac{m_1!}{k_1!} \cdots \frac{m_{p-1}!}{k_{p-1}!} \cdot (d_p - h)!$. □

Corollary 3.20. *In the same situation as Lemma 3.19, we have*

$$|\Pi_{I,I}(F)| \cdot |\Pi_{\bar{I},\bar{I}}(F)| \leq |\Pi_{J,J}(F)| \cdot |\Pi_{\bar{J},\bar{J}}(F)|.$$

Proof. By Lemma 3.18, we have

$$|\Pi_{I,I}(F)| \cdot |\Pi_{\bar{I},\bar{I}}(F)| = \frac{\lfloor \frac{m_1}{2} \rfloor! \lceil \frac{m_1}{2} \rceil! \cdots \lfloor \frac{m_r}{2} \rfloor! \lceil \frac{m_r}{2} \rceil!}{\lfloor \frac{k_1}{2} \rfloor! \lceil \frac{k_1}{2} \rceil! \cdots \lfloor \frac{k_{r-1}}{2} \rfloor! \lceil \frac{k_{r-1}}{2} \rceil!}. \tag{3.21}$$

Now apply Proposition 3.11 to

$$\frac{\lfloor \frac{m_i}{2} \rfloor! \lceil \frac{m_i}{2} \rceil!}{\lfloor \frac{k_i}{2} \rfloor! \lceil \frac{k_i}{2} \rceil!}$$

for $i = 1, \ldots, p-1, p+1, \ldots, r-1$ and apply Proposition 3.5 to $\lfloor \frac{m_p}{2} \rfloor! \lceil \frac{m_p}{2} \rceil!$ and $\lfloor \frac{m_r}{2} \rfloor! \lceil \frac{m_r}{2} \rceil!$, using the sums $m_p = (h - c_p + 1) + (d_p - h)$, $m_r = m_r + 0$. We have that (3.21) is at most

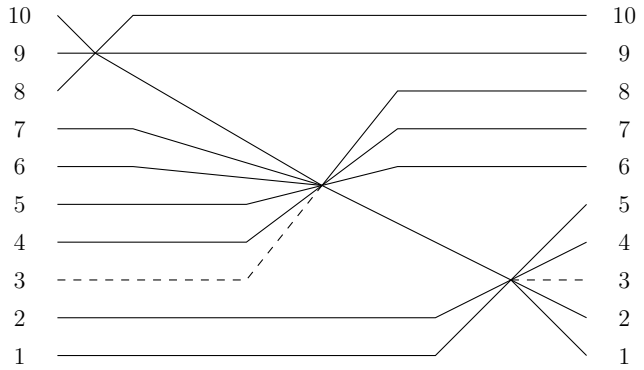
$$\frac{(h - c_p + 1)!(d_p - h)!}{\left\lfloor \frac{k_p}{2} \right\rfloor! \left\lceil \frac{k_p}{2} \right\rceil!} \cdot \frac{m_1!}{k_1!} \cdots \frac{m_{p-1}!}{k_{p-1}!} \frac{m_{p+1}!}{k_{p+1}!} \cdots \frac{m_{r-1}!}{k_{r-1}!} \cdot m_r!.$$

By Lemma 3.19, this equals to

$$\frac{|\Pi_{J,J}(F)| \cdot |\Pi_{\bar{J},\bar{J}}(F)|}{\lfloor \frac{k_p}{2} \rfloor! \lceil \frac{k_p}{2} \rceil!}.$$

□

Now consider the variation of (3.17) which counts paths in $\Pi_{[n]\setminus\{i\},[n]\setminus\{i\}}(F)$ when F is a descending star network of order n. For example, when $F = F_{[8,10]} \bullet F_{[3,8]} \bullet F_{[1,5]}$ and $i = 3$, these are the paths covering the solid edges in



We see that $|\Pi_{[n]\setminus\{i\},[n]\setminus\{i\}}(F)| = \frac{3!5!4!}{1!2!}$.

More generally, we have the following.

Lemma 3.21. *Let $F = F_{[c_1,d_1]} \bullet \cdots \bullet F_{[c_r,d_r]}$ be a descending star network with m_i, k_i defined as* (3.16)*. Let $\{i\} \in [c_p, d_p] \cap [c_{p+1}, d_{p+1}] \cap \cdots \cap [c_q, d_q]$ for some $p, q \in [1, r]$. Then, we have*

$$|\Pi_{[n]\setminus\{i\},[n]\setminus\{i\}}(F)| \cdot |\Pi_{\{i\},\{i\}}(F)| = \frac{m_1! \cdots m_{p-1}!(m_p - 1)! \cdots (m_q - 1)! m_{q+1}! \cdots m_r!}{k_1! \cdots k_{p-1}!(k_p - 1)! \cdots (k_{q-1} - 1)! k_q! \cdots k_{r-1}!}.$$

Proof. Note that $|\Pi_{\{i\},\{i\}}(F)| = 1$, so we just need to compute $|\Pi_{J,J}(F)|$. Delete source i and sink i from F, and delete all edges incident upon source i or sink i from F. Then the resulting graph is a new descending star network with $n-1$ sources and $n-1$ sinks. Re-label sources and sinks by from bottom to top by $\{1, \cdots, n-1\}$, we obtain the desired expression by Lemma 3.17. □

4. Merging stars in a star network

We introduce some elementary permanent inequalities that will be useful later.

Proposition 4.1. *Let A and B be two $n \times n$ matrices with $0 \le a_{ij} \le b_{ij}$ for all i, j. Then* $\operatorname{per}(A) \le \operatorname{per}(B)$*. In particular, for $L \subseteq [n]$, we have*

$$\operatorname{per}(A_{L,L}) \operatorname{per}(A_{\bar{L}\bar{L}}) \le \operatorname{per}(B_{L,L}) \operatorname{per}(B_{\bar{L}\bar{L}}). \tag{4.1}$$

Proof. For all $w \in \mathfrak{S}_n$, we have $a_{1,w_1} \cdots a_{n,w_n} \le b_{1,w_1} \cdots b_{n,w_n}$, and $\operatorname{per}(A) \le \operatorname{per}(B)$. □

Observe that for a general descending star network F, there may be multiple stars $F_{[c_p,d_p]}$ with $[c_p, d_p] \cap [h, h+1] \neq \emptyset$. We now define a function that can modify the star network F so that the resulting star network satisfies the conditions in Lemma 3.19. For each number $h \in [n]$, define the *h-merge function* $\phi_h : \mathcal{D}_n \to \mathcal{D}_n$ as follows.

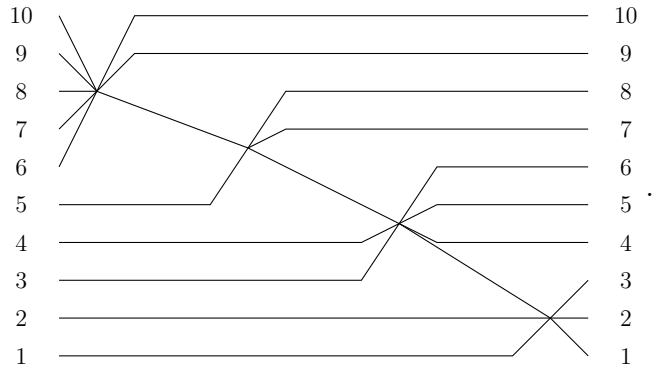
(1) Given $F = F_{[c_1,d_1]} \bullet \cdots \bullet F_{[c_r,d_r]}$, define

$$p = \max\{k \mid [c_k, d_k] \cap [h, h+1] \neq \emptyset\},$$
$$q = \min\{k \mid [c_k, d_k] \cap [h, h+1] \neq \emptyset\}.$$

(2) Define $\phi_h(F) = F_{[c_1,d_1]} \bullet \cdots \bullet F_{[c_{q-1},d_{q-1}]} \bullet F_{[c_p,d_q]} \bullet F_{[c_{p+1},d_{p+1}]} \bullet \cdots \bullet F_{[c_r,d_r]}$.

For example, consider

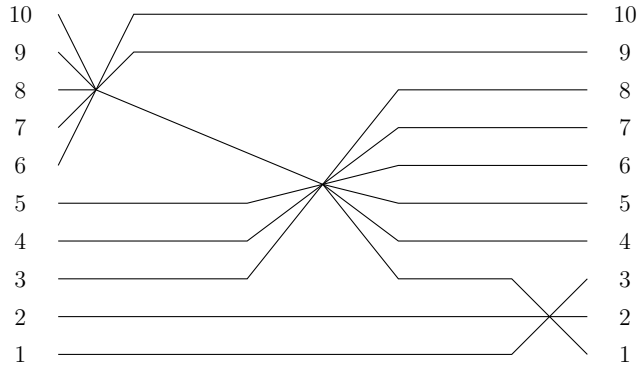
$$F = F_{[6,10]} \bullet F_{[5,8]} \bullet F_{[3,6]} \bullet F_{[1,3]} =$$



.

Observe that we have $\phi_1(F) = \phi_9(F) = \phi_{10}(F) = F$, since $[1,2]$, $[9,10]$, $[10,11]$ each intersects only one of the four given intervals. On the other hand, we have

$$\phi_4(F) = F_{[6,10]} \bullet F_{[3,8]} \bullet F_{[1,3]} =$$



,

since the intersection of $[4,5]$ with $[3,6]$ and $[5,8]$ leads to the merging of these two intervals into $[3,8]$. Similarly, we have

$$\phi_2(F) = \phi_3(F) = F_{[6,10]} \bullet F_{[5,8]} \bullet F_{[1,6]},$$
$$\phi_5(F) = \phi_6(F) = F_{[3,10]} \bullet F_{[1,3]},$$
$$\phi_7(F) = \phi_8(F) = F_{[5,10]} \bullet F_{[3,6]} \bullet F_{[1,3]}.$$

The path matrices of F and $\phi_h(F)$ have the following permanent inequality.

Corollary 4.2. *Let F be a descending star network of order n and fix $h \in [n]$. Let A be the path matrix of F and B be the path matrix of $\phi_h(F)$. Then for all $L \subseteq [n]$ we have*

$$\operatorname{per}(A_{L,L}) \operatorname{per}(A_{\bar{L}\bar{L}}) \leq \operatorname{per}(B_{L,L}) \operatorname{per}(B_{\bar{L}\bar{L}}).$$

Proof. Observe that if there is a path from source i to sink j in F, then there is also a path from the same source and sink in $\phi_h(F)$. Therefore, $0 \leq a_{ij} \leq b_{ij}$ for all $i, j \in [n]$. Then we have the desired inequality by Proposition 4.1. □

When $L = [h]$, Corollary 4.2 has a special case where the inequality becomes an equality.

Proposition 4.3. *Fix $h \in [n]$, and let F be a descending star network of order n with path matrix A. Let B be the path matrix of the descending star network $\phi_h(F)$. Then we have*

(i) $A_{[h],[h]} = B_{[h],[h]}$ and $A_{[h+1,n],[h+1,n]} = B_{[h+1,n],[h+1,n]}$.

(ii) $|\Pi_{[h],[h]}(F)| \cdot |\Pi_{[h+1,n],[h+1,n]}(F)| = |\Pi_{[h],[h]}(\phi_h(F))| \cdot |\Pi_{[h+1,n],[h+1,n]}(\phi_h(F))|$.

Proof. (i) First, we show that $A_{[1,h],[1,h]} = B_{[1,h],[1,h]}$.

Case 1: Let $i > j \in [1,h]$. If $a_{ij} = 1$, then, source i and sink j belong to the same component of F. It's easy to see that they also belong to the same component of $\phi_h(F)$. Hence, $b_{ij} = 1$. If $a_{ij} = 0$, then source i and sink j do not belong to the same component of F. By the definition of ϕ_h, i and j do not belong to the same component of $\phi_h(F)$. Hence, $b_{ij} = 0$.

Case 2: Let $i \leq j \in [1,h]$. If $a_{ij} = 1$, there is a path from source i to sink j in F. So, there exists an index m such that $i, j \in [c_m, d_m]$. Since every interval appearing in $\phi_h(F)$ is a union of some intervals of F, i, j are in an interval of $\phi_h(F)$. Hence, $b_{ij} = 1$. If $a_{ij} = 0$, then no intervals contain both i and j. In particular, no intervals contain both h and i, then all intervals appearing in $\phi_h(F)$ containing i are equal to intervals containing i in F. None of these intervals also contain j, so $b_{ij} = 0$.

Therefore, $A_{[1,h],[1,h]} = B_{[1,h],[1,h]}$. Similarly, we have $A_{[h+1,n],[h+1,n]} = B_{[h+1,n],[h+1,n]}$.

(ii) By Observation 3.15 and part (i), we have

$$|\Pi_{[1,h],[1,h]}(F)| = \operatorname{per}(A_{[1,h],[1,h]}) = \operatorname{per}(B_{[1,h],[1,h]}) = |\Pi_{[1,h],[1,h]}(\phi_h(F))|,$$
$$|\Pi_{[h+1,n],[h+1,n]}(F)| = \operatorname{per}(A_{[h+1,n],[h+1,n]}) = \operatorname{per}(B_{[h+1,n],[h+1,n]}) = |\Pi_{[h+1,n],[h+1,n]}(\phi_h(F))|.$$

□

In the previous example $F = F_{[6,10]} \bullet F_{[5,8]} \bullet F_{[3,6]} \bullet F_{[1,3]}$ and $\phi_4(F) = F_{[6,10]} \bullet F_{[3,8]} \bullet F_{[1,3]}$, one can see that

$$|\Pi_{[1,4],[1,4]}(F)| = \operatorname{per}(A_{[1,4],[1,4]}) = |\Pi_{[1,4],[1,4]}(\phi_4(F))| = \operatorname{per}(B_{[1,4],[1,4]}) = 12,$$
$$|\Pi_{[5,10],[5,10]}(F)| = \operatorname{per}(A_{[5,10],[5,10]}) = |\Pi_{[5,10],[5,10]}(\phi_4(F))| = \operatorname{per}(B_{[5,10],[5,10]}) = 480,$$

where

$$A = \begin{bmatrix} 1&1&1&0&0&0&0&0&0&0\\ 1&1&1&0&0&0&0&0&0&0\\ 1&1&1&1&1&1&0&0&0&0\\ 1&1&1&1&1&1&0&0&0&0\\ 1&1&1&1&1&1&1&1&0&0\\ 1&1&1&1&1&1&1&1&1&1\\ 1&1&1&1&1&1&1&1&1&1\\ 1&1&1&1&1&1&1&1&1&1\\ 1&1&1&1&1&1&1&1&1&1\\ 1&1&1&1&1&1&1&1&1&1 \end{bmatrix}, \quad B = \begin{bmatrix} 1&1&1&0&0&0&0&0&0&0\\ 1&1&1&0&0&0&0&0&0&0\\ 1&1&1&1&1&1&\mathbf{1}&\mathbf{1}&0&0\\ 1&1&1&1&1&1&\mathbf{1}&\mathbf{1}&0&0\\ 1&1&1&1&1&1&1&1&0&0\\ 1&1&1&1&1&1&1&1&1&1\\ 1&1&1&1&1&1&1&1&1&1\\ 1&1&1&1&1&1&1&1&1&1\\ 1&1&1&1&1&1&1&1&1&1\\ 1&1&1&1&1&1&1&1&1&1 \end{bmatrix}.$$

5. Main results

Now we can state and prove our main result about products of permanents.

Theorem 5.1. *Let P be an n-element unit interval order labeled as in Algorithm 2.2 and let A be its antiadjacency matrix. Fix $h \in [n]$ and define sets*

$$I = [h], \qquad J = [n] \cap 2\mathbb{Z}.$$

Then we have

$$\operatorname{per}(A_{I,I}) \operatorname{per}(A_{\bar{I},\bar{I}}) \geq \operatorname{per}(A_{J,J}) \operatorname{per}(A_{\bar{J},\bar{J}}). \tag{5.1}$$

Proof. Let

$$F = F_{[c_1,d_1]} \bullet \cdots \bullet F_{[c_r,d_r]} \tag{5.2}$$

be the descending star network corresponding to P as in Algorithm 2.5 and let B be the path matrix of F. By Proposition 2.11 we have $B = \psi(A)$. By Corollary 2.10 the inequality (5.1) holds for A if and only if it holds for B. By Corollary 3.16 we have

$$\begin{aligned} \operatorname{per}(B_{I,I}) \operatorname{per}(B_{\bar{I},\bar{I}}) &= |\Pi_{I,I}(F)| \cdot |\Pi_{\bar{I},\bar{I}}(F)|, \\ \operatorname{per}(B_{J,J}) \operatorname{per}(B_{\bar{J},\bar{J}}) &= |\Pi_{J,J}(F)| \cdot |\Pi_{\bar{J},\bar{J}}(F)|, \end{aligned} \tag{5.3}$$

and it suffices to show that we have $|\Pi_{I,I}(F)| \cdot |\Pi_{\bar{I},\bar{I}}(F)| \geq |\Pi_{J,J}(F)| \cdot |\Pi_{\bar{J},\bar{J}}(F)|$.

Suppose first that no interval $[c_i, d_i]$ in (5.2) contains $[h, h+1]$ as a subinterval. Then each source-to-sink path in F has its source and sink both in $[h]$ or both in $[h+1, n]$. Thus we have $|\Pi_{I,I}(F)| \cdot |\Pi_{\bar{I},\bar{I}}(F)| = |\Pi_{[n],[n]}(F)|$. By Lemma 3.17 this cardinality is

$$\frac{m_1! \cdots m_r!}{k_1! \cdots k_{r-1}!}, \tag{5.4}$$

where $m_i := \operatorname{size}(F_{[c_i,d_i]})$, and $k_i := \operatorname{overlap}(F_{[c_i,d_i]}, F_{[c_{i+1},d_{i+1}]})$. But by Propositions 3.5 and 3.11 this is at least

$$\frac{\lfloor \frac{m_1}{2} \rfloor! \lceil \frac{m_1}{2} \rceil! \cdots \lfloor \frac{m_r}{2} \rfloor! \lceil \frac{m_r}{2} \rceil!}{\lfloor \frac{k_1}{2} \rfloor! \lceil \frac{k_1}{2} \rceil! \cdots \lfloor \frac{k_{r-1}}{2} \rfloor! \lceil \frac{k_{r-1}}{2} \rceil!},$$

which by Lemma 3.18 equals $|\Pi_{J,J}(F)| \cdot |\Pi_{\bar{J},\bar{J}}(F)|$.

Now suppose that some interval $[c_i, d_i]$ in (5.2) does contain $[h, h+1]$ as a subinterval. Then we do not have the factorization above (5.4), but by Proposition 4.3 we have

$$|\Pi_{I,I}(F)| \cdot |\Pi_{\bar{I},\bar{I}}(F)| = |\Pi_{I,I}(\phi_h(F))| \cdot |\Pi_{\bar{I},\bar{I}}(\phi_h(F))|.$$

By Corollary 3.20, this is at least $|\Pi_{J,J}(\phi_h(F))| \cdot |\Pi_{\bar{J},\bar{J}}(\phi_h(F))|$, and by Corollary 4.2 this is at least $|\Pi_{J,J}(F)| \cdot |\Pi_{\bar{J},\bar{J}}(F)|$. □

Theorem 5.2. *Let P be an n-element unit interval order labeled as in Algorithm 2.2 and let A be its antiadjacency matrix. Let $J = [n] \cap 2\mathbb{Z}$. Then for any $i \in [n]$, we have*

$$\operatorname{per}(A_{[n]\setminus\{i\},[n]\setminus\{i\}}) \operatorname{per}(A_{\{i\},\{i\}}) \geq \operatorname{per}(A_{J,J}) \operatorname{per}(A_{\bar{J},\bar{J}}). \tag{5.5}$$

Proof. Let

$$F = F_{[c_1,d_1]} \bullet \cdots \bullet F_{[c_r,d_r]}$$

be the descending star network corresponding to P as in Algorithm 2.5 and let B be the path matrix of F. By Proposition 2.11 we have $B = \psi(A)$. By Corollary 2.10 the inequality (5.5) holds for A if and only if it holds for B. By Corollary 3.16 we have

$$\begin{aligned} \operatorname{per}(B_{[n]\setminus\{i\},[n]\setminus\{i\}}) \operatorname{per}(B_{\{i\},\{i\}}) &= |\Pi_{[n]\setminus\{i\},[n]\setminus\{i\}}(F)| \cdot |\Pi_{\{i\},\{i\}}(F)|, \\ \operatorname{per}(B_{J,J}) \operatorname{per}(B_{\bar{J},\bar{J}}) &= |\Pi_{J,J}(F)| \cdot |\Pi_{\bar{J},\bar{J}}(F)|, \end{aligned} \tag{5.6}$$

and it suffices to show that we have

$$|\Pi_{[n]\setminus\{i\},[n]\setminus\{i\}}(F)| \cdot |\Pi_{\{i\},\{i\}}(F)| \geq |\Pi_{J,J}(F)| \cdot |\Pi_{\bar{J},\bar{J}}(F)|. \tag{5.7}$$

Suppose that $i \notin [c_p, d_p]$ for all $p \in [1, r]$. Then, in the descending star network F, the path from source i to sink i is a simple (directed) edge. Hence, the left hand side of (5.7) is

$$|\Pi_{[n]\setminus\{i\},[n]\setminus\{i\}}(F)| = |\Pi_{[n],[n]}(F)|.$$

Since every path family with odd sources going to odd sinks and even sources going to even sinks belongs to $\Pi_{[n],[n]}(F)$, this is at least

$$|\Pi_{J,J}(F)| \cdot |\Pi_{\bar{J},\bar{J}}(F)|,$$

and (5.7) is true.

On the other hand, suppose that $i \in [c_p, d_p] \cap [c_{p+1}, d_{p+1}] \cap \cdots \cap [c_q, d_q]$ for some $p, q \in [1, r]$. Then, by Lemma 3.21, the left-hand side of (5.7) is

$$\frac{m_1! \cdots m_{p-1}!(m_p - 1)! \cdots (m_{q-1} - 1)!(m_q - 1)! m_{q+1}! \cdots m_r!}{k_1! \cdots k_{p-1}!(k_p - 1)! \cdots (k_{q-1} - 1)! k_q! \cdots k_{r-1}!}.$$

Note that $m_j \geq k_j$ for $j = 1, \ldots, q-1$ and $m_{j+1} \geq k_j$ for $j = q, \ldots r-1$. By Propositions 3.11, 3.12, the above fraction is at least

$$\frac{\lfloor \frac{m_1}{2} \rfloor! \lceil \frac{m_1}{2} \rceil!}{\lfloor \frac{k_1}{2} \rfloor! \lceil \frac{k_1}{2} \rceil!} \cdots \frac{\lfloor \frac{m_{q-1}}{2} \rfloor! \lceil \frac{m_{q-1}}{2} \rceil!}{\lfloor \frac{k_{q-1}}{2} \rfloor! \lceil \frac{k_{q-1}}{2} \rceil!} (m_q - 1)! \frac{\lfloor \frac{m_{q+1}}{2} \rfloor! \lceil \frac{m_{q+1}}{2} \rceil!}{\lfloor \frac{k_q}{2} \rfloor! \lceil \frac{k_q}{2} \rceil!} \cdots \frac{\lfloor \frac{m_r}{2} \rfloor! \lceil \frac{m_r}{2} \rceil!}{\lfloor \frac{k_{r-1}}{2} \rfloor! \lceil \frac{k_{r-1}}{2} \rceil!}.$$

Applying Proposition 3.5 to $(m_q - 1)! = (m_q - 1)! \cdot 1!$, we see that this is at least

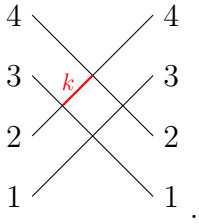
$$\frac{\lfloor \frac{m_1}{2} \rfloor! \lceil \frac{m_1}{2} \rceil! \cdots \lfloor \frac{m_r}{2} \rfloor! \lceil \frac{m_r}{2} \rceil!}{\lfloor \frac{k_1}{2} \rfloor! \lceil \frac{k_1}{2} \rceil! \cdots \lfloor \frac{k_{r-1}}{2} \rfloor! \lceil \frac{k_{r-1}}{2} \rceil!},$$

which is $|\Pi_{J,J}(F)| \cdot |\Pi_{\bar{J},\bar{J}}(F)|$, by Lemma 3.18. Again (5.7) is true. □

One may guess that we can extend Theorem 5.2 to all TNN matrices. However, there is a counterexample when $n = 4$ and $i = 3$. Let

$$M(k) = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1+k & 1+k & k \\ 1 & 1+k & 1+k & k \\ 0 & 1 & 1 & 1 \end{bmatrix}, \text{ with } k \geq 0.$$

Note that $M(k)$ is TNN since it is the path matrix of the weighted planar network



Then, we have

$$\operatorname{per}(M(k)_{124,124}) \operatorname{per}(M(k)_{3,3}) - \operatorname{per}(M(k)_{13,13}) \operatorname{per}(M(k)_{24,24}) = -k.$$

Therefore, (5.5) is not always true for all TNN matrices.

6. A Generalization on J

A natural generalization of a product of two permanents indexed by even or odd integers is a product of s permanents indexed by integers which are equivalent modulo s. Define

$$[c, d]_r^s = \{x \in [c, d] : \ x \equiv r \pmod{s}\}.$$

For example, $[5,12]_0^3 = \{6,9,12\}$, $[5,12]_1^3 = \{7,10\}$, and $[5,12]_2^3 = \{5,8,11\}$. Naturally, the cardinalities of these sets differ by at most one, and sum to $|[5,12]| = 8$. More generally, we have the following elementary number theoretic result.

Observation 6.1. *For $0 < s \leq n$ and $[c,d] \subseteq [1,n]$, the cardinalities of $[c,d]_0^s, \ldots, [c,d]_{s-1}^s$ differ by at most one.*

Proof. Use integer division to write $d - c + 1 = qs + r$ with $r < s$. Then r intervals have cardinality $q+1$ and $s - r$ intervals have cardinality q. □

Observe that $q = \left\lfloor \frac{d-c+1}{s} \right\rfloor$ and $q+1 = \left\lceil \frac{d-c+1}{s} \right\rceil$. This leads to the following generalization of Proposition 3.11.

Proposition 6.2. *Fix $s, m, k \in \mathbb{N}$ with $k \leq m$ and define integers $k_0, \ldots, k_{s-1}, m_0, \ldots, m_{s-1}$ by $k_i = |[1,k]_i^s|$, $m_i = |[1,m]_i^s|$. These numbers satisfy*

$$\frac{m!}{k!} \geq \frac{m_0! m_1! \cdots m_{s-1}!}{k_0! k_1! \cdots k_{s-1}!}. \tag{6.1}$$

Proof. The inequality (6.1) is equivalent to

$$\binom{m}{m_0, m_1, \ldots, m_{s-1}} \geq \binom{k}{k_0, k_1, \ldots, k_{s-1}}. \tag{6.2}$$

By the multinomial analog of the Pascal's identity,

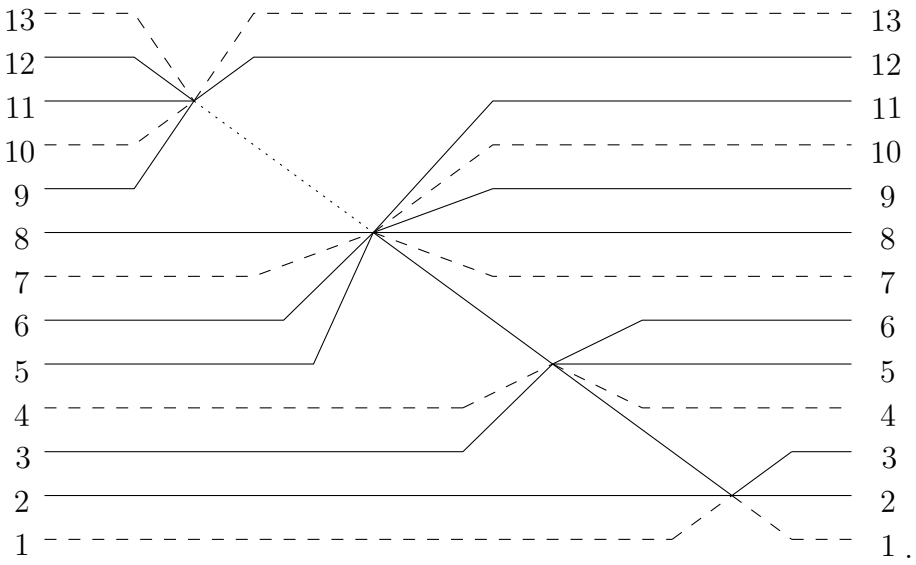
$$\binom{n}{a_1, \ldots, a_t} = \binom{n-1}{a_1 - 1, a_2, \ldots, a_t} + \binom{n-1}{a_1, a_2 - 1, \ldots, a_t} + \cdots + \binom{n-1}{a_1, \ldots, a_t - 1}, \tag{6.3}$$

we see that the left-hand side of (6.3) is greater than each term on the right-hand side of (6.3). It follows that for $b_1, \cdots, b_t \in \mathbb{N}$, with $b_i \leq a_i$ for $i = 1, \ldots, t$ and $b = b_1 + \cdots b_t$, we have

$$\binom{n}{a_1, \ldots, a_t} \geq \binom{n-b}{a_1 - b_1, a_2 - b_2, \ldots, a_t - b_t}.$$

The inequality (6.2) follows. □

Now consider the variation of (3.17) which counts paths in $\Pi_{[n],[n]}(F)$ having sources and sinks with only indices that are 1 modulo 3. For example, when $F = F_{[9,13]} \bullet F_{[5,11]} \bullet F_{[3,6]} \bullet F_{[1,3]}$, these are the paths covering the solid and dotted edges in



Then, we see that

$$\Pi_{\{1,4,7,10,13\},\{1,4,7,10,13\}}(F) = \frac{2!2!1!1!}{1!0!0!} = 4.$$

More generally, we have the following lemma.

Lemma 6.3. *Let $F = F_{[c_0,d_0]} \bullet \cdots \bullet F_{[c_t,d_t]}$ be a descending star network. Fix $s \in [n]$. For every $r \in [0, s-1]$, we have*

$$|\Pi_{[1,n]_r^s,[1,n]_r^s}(F)| = \frac{|[c_0,d_0]_r^s|! \cdot |[c_1,d_1]_r^s|! \cdots |[c_t,d_t]_r^s|!}{|[c_0,d_1]_r^s|! \cdot |[c_1,d_2]_r^s|! \cdots |[c_{t-1},d_t]_r^s|!}. \tag{6.4}$$

Proof. Observe that $|\Pi_{[1,n]_r^s,[1,n]_r^s}(F)|$ corresponds to a sub-star network with sources and sinks labeled in $[1,n]_r^s$. By Lemma 3.17, we have (6.4). □

We can now extend Theorem 5.1 to prove inequalities such as the following for 9×9 matrices:

$$\begin{aligned}\operatorname{per}(A) &\geq \operatorname{per}(A_{13579,13579}) \operatorname{per}(A_{2468,2468}) \\ &\geq \operatorname{per}(A_{147,147}) \operatorname{per}(A_{258,258}) \operatorname{per}(A_{369,369}) \\ &\geq \operatorname{per}(A_{159,159}) \operatorname{per}(A_{26,26}) \operatorname{per}(A_{37,37}) \operatorname{per}(A_{48,48}) \geq \cdots \\ &\geq \operatorname{per}(A_{19,19}) a_{22} \cdots a_{88} \\ &\geq a_{11} \cdots a_{99}.\end{aligned}$$

Theorem 6.4. *Let A be the antiadjacency matrix of an n-element unit interval order. Then for $s = 2, \ldots, n-1$, we have*

$$\operatorname{per}(A_{[1,n]_0^s,[1,n]_0^s}) \cdots \operatorname{per}(A_{[1,n]_{s-1}^s,[1,n]_{s-1}^s}) \geq \operatorname{per}(A_{[1,n]_0^{s+1},[1,n]_0^{s+1}}) \cdots \operatorname{per}(A_{[1,n]_s^{s+1},[1,n]_s^{s+1}}). \tag{6.5}$$

Proof. Let $F = F_{[c_0,d_0]} \bullet \cdots \bullet F_{[c_t,d_t]}$ be the descending star network corresponding to A, i.e., the descending star network of $\psi(A)$ from Proposition 2.11. Then by Corollary 3.16, the rth factor on the left-hand-side of (6.5) is $|\Pi_{[1,n]_r^s,[1,n]_r^s}(F)|$. By Lemma 6.3, this equals $|[c_t,d_t]_r^s|!$ times

$$\alpha_r := \frac{|[c_0,d_0]_r^s|!}{|[c_0,d_1]_r^s|!} \cdot \frac{[c_1,d_1]_r^s|!}{|[c_1,d_2]_r^s|!} \cdots \frac{|[c_{t-1},d_{t-1}]_r^s|!}{|[c_{t-1},d_t]_r^s|!}, \tag{6.6}$$

i.e., the left-hand-side of (6.5) equals

$$|[c_t, d_t]_0^s|!\alpha_0 \cdots |[c_t, d_t]_{s-1}^s|!\alpha_{s-1}. \tag{6.7}$$

Now fix $i \in [0, t-1]$ and collect the ratios $\frac{|[c_i,d_i]_r^s|!}{|[c_i,d_{i+1}]_r^s|!}$ for $r = 0, \ldots, s-1$. The product

$$\frac{|[c_i, d_i]_0^s|!}{|[c_i, d_{i+1}]_0^s|!} \cdots \frac{|[c_i, d_i]_{s-1}^s|!}{|[c_i, d_{i+1}]_{s-1}^s|!} \tag{6.8}$$

satisfies

$$\begin{aligned} |[c_i, d_i]_0^s| + \cdots + |[c_i, d_i]_{s-1}^s| &= d_i - c_i + 1, \\ |[c_i, d_{i+1}]_0^s| + \cdots + |[c_i, d_{i+1}]_{s-1}^s| &= d_{i+1} - c_i + 1, \end{aligned} \tag{6.9}$$

with terms within each sum differing from one another by at most 1, and

$$d_i - c_i + 1 \geq d_{i+1} - c_i + 1.$$

We similarly apply Corollary 3.16 to the rth factor on the right-hand-side of (6.5) and Lemma 6.3 to $|\Pi_{[1,n]_r^{s+1},[1,n]_r^{s+1}}(F)|$. This expression equals $|[c_t, d_t]_r^{s+1}|!$ times

$$\beta_r := \frac{|[c_0, d_0]_r^{s+1}|!}{|[c_0, d_1]_r^{s+1}|!} \cdot \frac{[c_1, d_1]_r^{s+1}|!}{|[c_1, d_2]_r^{s+1}|!} \cdots \frac{|[c_{t-1}, d_{t-1}]_r^{s+1}|!}{|[c_{t-1}, d_t]_r^{s+1}|!}, \tag{6.10}$$

i.e., the right-hand-side of (6.5) equals

$$|[c_t, d_t]_0^{s+1}|!\beta_0 \cdots |[c_t, d_t]_s^{s+1}|!\beta_s. \tag{6.11}$$

Again fixing $i \in [0, t-1]$ and collecting ratios $\frac{|[c_i,d_i]_r^{s+1}|!}{|[c_i,d_{i+1}]_r^{s+1}|!}$ for $r = 0, \ldots, s$, we obtain a product analogous to (6.8) in which numerator (denominator) cardinalities differ from one another by at most 1, and have sums matching those in (6.9). By Corollary 3.10, the two products satisfy

$$\frac{|[c_i, d_i]_0^s|!}{|[c_i, d_{i+1}]_0^s|!} \cdots \frac{|[c_i, d_i]_{s-1}^s|!}{|[c_i, d_{i+1}]_{s-1}^s|!} \geq \frac{|[c_i, d_i]_0^{s+1}|!}{|[c_i, d_{i+1}]_0^{s+1}|!} \cdots \frac{|[c_i, d_i]_s^{s+1}|!}{|[c_i, d_{i+1}]_s^{s+1}|!}, \tag{6.12}$$

and therefore we have

$$\alpha_0 \cdots \alpha_{s-1} \geq \beta_0 \cdots \beta_s. \tag{6.13}$$

Now consider the cardinalities of the intervals indexed by t. By Corollary 3.2 we have

$$|[c_t, d_t]_0^s|! \cdots |[c_t, d_t]_{s-1}^s|! \geq |[c_t, d_t]_0^{s+1}|! \cdots |[c_t, d_t]_s^{s+1}|!. \tag{6.14}$$

Combining (6.13) and (6.14), we have that (6.7) is at least (6.11), as desired. □

Combine Theorem 6.4 and Theorem 5.1, and we have the corollary.

Corollary 6.5. *Let A be an $n \times n$ antiadjacency matrix of a unit interval order. Fix $h \in [n]$. Then, for $s = 2, \ldots, n-1$, we have*

$$\operatorname{per}(A_{[1,h],[1,h]}) \operatorname{per}(A_{[h+1,n],[h+1,n]}) \geq \operatorname{per}(A_{[1,n]_0^s,[1,n]_0^s}) \cdots \operatorname{per}(A_{[1,n]_{s-1}^s,[1,n]_{s-1}^s}).$$

Combine Theorem 6.4 and Theorem 5.2, and we have the corollary.

Corollary 6.6. *Let A be an $n \times n$ antiadjacency matrix of a unit interval order. Fix $i \in [n]$. Then, for $s = 2, \ldots, n-1$, we have*

$$\operatorname{per}(A_{[n]\setminus\{i\},[n]\setminus\{i\}}) \operatorname{per}(A_{\{i\},\{i\}}) \geq \operatorname{per}(A_{[1,n]_0^s,[1,n]_0^s}) \cdots \operatorname{per}(A_{[1,n]_{s-1}^s,[1,n]_{s-1}^s}).$$

7. A Generalization to all TNN matrices

In this section, we aim to show that (1.9) holds for $I = [1, \lfloor \frac{n}{2} \rfloor]$.

A polynomial $p(x_{11}, x_{12}, \cdots, x_{nn})$ in n^2 variables is called totally non-negative (TNN) if it satisfies

$$p(A) := p(a_{11}, a_{12}, \cdots, a_{nn}) \geq 0 \tag{7.1}$$

for each $n \times n$ totally non-negative matrix $A = (a_{ij})$. Therefore, (1.9) is true if and only if the polynomial

$$\operatorname{per}(X_{[1,\lfloor \frac{n}{2} \rfloor],[1,\lfloor \frac{n}{2} \rfloor]}) \operatorname{per}(X_{[\lfloor \frac{n}{2} \rfloor+1,n],[\lfloor \frac{n}{2} \rfloor+1,n]}) - \operatorname{per}(X_{[n]\cap 2\mathbb{Z},[n]\cap 2\mathbb{Z}}) \operatorname{per}(X_{[n]\setminus 2\mathbb{Z},[n]\setminus 2\mathbb{Z}}) \tag{7.2}$$

is TNN.

Drake, Gerrish and Skandera [5] gave a necessary and sufficient condition for a polynomial which is a difference of "permutation monomials" to be TNN.

Theorem 7.1 (Drake, Gerrish, Skandera)**.** *Let w and u be two permutations in $\mathfrak{S}_n$. Then w is less than or equal to u in the (strong) Bruhat order if and only if the polynomial*

$$x_{1,w(1)} \cdots x_{n,w(n)} - x_{1,u(1)} \cdots x_{n,u(n)}$$

is totally non-negative.

Let $t = \lfloor \frac{n}{2} \rfloor$. Observe that the positive term $\operatorname{per}(X_{\lfloor \frac{n}{2} \rfloor,\lfloor \frac{n}{2} \rfloor}) \operatorname{per}(X_{\lceil \frac{n}{2} \rceil,\lceil \frac{n}{2} \rceil})$ contains monomials

$$x_{1,w_1} \cdots x_{t,w_t} x_{t+1,w_{t+1}} \cdots x_{n,w_n} \quad \text{such that}$$

$$w_1 \cdots w_t \text{ permutes } \{1, \ldots, t\},$$
$$w_{t+1} \cdots w_n \text{ permutes } \{t+1, \ldots, n\},$$

or, in other words, using the Young subgroup,

$$w = w_1 \cdots w_n \in \mathfrak{S}_{1,\ldots,t} \times \mathfrak{S}_{t+1,\ldots,n}.$$

For convenience, we set

$$\begin{aligned} &\mathfrak{A}_n = \mathfrak{S}_{1,\ldots,t} \times \mathfrak{S}_{t+1,\ldots,n} \\ &= \{w \in \mathfrak{S}_n \mid w_1 \cdots w_t \text{ permutes } \{1, \ldots, t\} \text{ and } w_{t+1} \cdots w_n \text{ permutes } \{t+1, \ldots, n\}\}. \end{aligned} \tag{7.3}$$

Since $t = \lfloor \frac{n}{2} \rfloor$, it is straightforward to see that $|\mathfrak{A}_n| = t!(n-t)! = \lfloor \frac{n}{2} \rfloor! \lceil \frac{n}{2} \rceil!$.

For example, $\mathfrak{A}_4 = \{1234, 1243, 2134, 2143\}$, and $|\mathfrak{A}_4| = 4$.

The negative term $\operatorname{per}(X_{[n]\cap 2\mathbb{Z},[n]\cap 2\mathbb{Z}}) \operatorname{per}(X_{[n]\setminus 2\mathbb{Z},[n]\setminus 2\mathbb{Z}})$ contains monomials

$$x_{1,u_1} \cdots x_{n,u_n} \quad \text{such that}$$

$$u_1 u_3 u_5 \cdots \text{ permutes } \{1, 3, 5, \cdots\} = [n] \setminus 2\mathbb{Z},$$
$$u_2 u_4 u_6 \cdots \text{ permutes } \{2, 4, 6, \cdots\} = [n] \cap 2\mathbb{Z}.$$

We call the set that consists of $u = u_1 \cdots u_n$ satisfying the above conditions $\mathfrak{B}_n$, i.e.,

$$\mathfrak{B}_n = \{u \in \mathfrak{S}_n \mid u_1 u_2 \cdots u_n \text{ takes odd and even integers alternately and } u_1 \text{ is odd}\} \tag{7.4}$$

In the literature [26], $\mathfrak{B}_n$ is also referred to as the set of parity alternating permutations

$$\{w = w_1 \ldots w_n \in \mathfrak{S}_n \,|\, w_1 \text{ is odd and } w_i + w_{i+1} \text{ is odd for all } i = 1, \ldots n-1\}.$$

It is easy to see that $|\mathfrak{B}_n| = \lfloor \frac{n}{2} \rfloor ! \lceil \frac{n}{2} \rceil !$. Then, the polynomial (7.2) has this number of positive terms (on the left) and negative terms (on the right). Therefore, if we can find a bijective map $f_n : \mathfrak{A}_n \to \mathfrak{B}_n$, such that $w \leq f_n(w)$ for all $w \in \mathfrak{A}_n$, by Theorem 7.1, the polynomial (7.2) is TNN.

Next, we introduce a few necessary maps that will be helpful.

For $p, q \in [n]$, define an insertion map ins_p and an insertion-swapping map inss_q (when $n+1-q$ is even) given by

$$\begin{aligned} \mathrm{ins}_p &: \mathfrak{S}_n \to \mathfrak{S}_{n+1}, \\ w = w_1 \cdots w_p \cdots w_n &\mapsto w_1 \cdots w_{p-1}(n+1)w_p \cdots w_n, \\ \mathrm{inss}_q &: \mathfrak{S}_n \to \mathfrak{S}_{n+1}, \\ w = w_1 \cdots w_q \cdots w_n &\mapsto w_1 \cdots w_{q-1}(n+1)w_{q+1}w_q w_{q+3} w_{q+2} \cdots w_n w_{n-1}. \end{aligned}$$

For example, let $w = 216435$. Then,

$$\mathrm{ins}_3(w) = 2176435, \quad \mathrm{inss}_3(w) = 2174653,$$

but $\mathrm{inss}_4(w)$ is not well-defined since $n+1-q = 7-4 = 3$ is not even.

Next, we introduce a straightforward property of the insertion map.

Lemma 7.2. *Let $v, w \in \mathfrak{S}_n$, and $v \leq w$ in the Bruhat order. If $j \leq i$ as integers, then $\mathrm{ins}_i(v) \leq \mathrm{ins}_j(w)$.*

Proof. $\mathrm{ins}_i(v) \leq \mathrm{ins}_i(w) \leq \mathrm{ins}_j(w)$. □

When n is even, we can build the set $\mathfrak{A}_{n+1}$ from $\mathfrak{A}_n$, by applying the insertion map.

Observation 7.3. *Fix $n = 2k$ and let $\mathfrak{A}_n$ be defined as in (7.3). Then, we have*

$$\mathfrak{A}_{n+1} = \{\mathrm{ins}_i(a) \mid i \in [k+1, n+1] \text{ and } a \in \mathfrak{A}_n\}.$$

Proof. Let $a \in \mathfrak{A}_n$. It is easy to see that the first k letters of $\mathrm{ins}_i(a)$ with $i \in [k+1, n+1]$ are numbers in $[1, k]$. Therefore,

$$\mathfrak{A}_{n+1} \supset \{\mathrm{ins}_i(a) \mid i \in [k+1, n+1] \text{ and } a \in \mathfrak{A}_n\}.$$

On the other hand, let $w = w_1 \cdots w_{n+1} \in \mathfrak{A}_{n+1}$ with $w_j = n+1$. Then, $j \in [k+1, n+1]$ by definition. Hence, the permutation $\overline{w} = \overline{w}_1 \cdots \overline{w}_n = w_1 \cdots w_{j-1} w_{j+1} \cdots w_{n+1} \in \mathfrak{A}_n$. Thus, $\mathrm{ins}_j(\overline{w}) = w$, which gives

$$\mathfrak{A}_{n+1} \subset \{\mathrm{ins}_i(a) \mid i \in [k+1, n+1] \text{ and } a \in \mathfrak{A}_n\}.$$

□

However, this construction of $\mathfrak{A}_{n+1}$ from $\mathfrak{A}_n$ does not hold when n is odd. For example, consider $321654 \in \mathfrak{A}_6$. Removing 6, we get 32154. However, 32154 is not in $\mathfrak{A}_5$ since 3 is at position 1. So we need a new set to build $\mathfrak{A}_6$, which we now define.

Let $\tilde{t} = \lceil \frac{n}{2} \rceil$. Define

$$\begin{aligned} \tilde{\mathfrak{A}}_n &= \mathfrak{S}_{1,\ldots,\tilde{t}} \times \mathfrak{S}_{\tilde{t}+1,\ldots,n} \\ &= \{w \in \mathfrak{S}_n \mid w_1 \cdots w_{\tilde{t}} \text{ permutes } \{1, \ldots, \tilde{t}\} \text{ and } w_{\tilde{t}+1} \cdots w_n \text{ permutes } \{\tilde{t}+1, \ldots, n\}\}. \end{aligned} \tag{7.5}$$

Now, let's look at the example 321654 again. Removing 6, we get 32154 which is in $\tilde{\mathfrak{A}}_5$. Therefore, we can build the set $\mathfrak{A}_{n+1}$ from $\tilde{\mathfrak{A}}_n$, by applying the insertion maps.

Observation 7.4. *Fix an odd integer $n = 2k+1$ and let $\tilde{\mathfrak{A}}_n$ be defined as in* (7.5)*. Then, we have*

$$\mathfrak{A}_{n+1} = \{\mathrm{ins}_i(\tilde{a}) \mid i \in [k+2, n+1] \text{ and } \tilde{a} \in \tilde{\mathfrak{A}}_n\}.$$

Proof. The proof is similar to the proof of Observation 7.3. □

Similar to $\mathfrak{A}_n$, we can build $\mathfrak{B}_{n+1}$ from $\mathfrak{B}_n$ by applying insertion-swapping maps.

Observation 7.5. *Let $\mathfrak{B}_n$ be defined as in* (7.4)*. Then, we have*

$$\mathfrak{B}_{n+1} = \begin{cases} \{\mathrm{inss}_j(b) \mid j \in [1, n+1] \setminus 2\mathbb{Z} \text{ and } b \in \mathfrak{B}_n\} & \text{when } n \text{ is even,} \\ \{\mathrm{inss}_j(b) \mid j \in [1, n+1] \cap 2\mathbb{Z} \text{ and } b \in \mathfrak{B}_n\} & \text{when } n \text{ is odd.} \end{cases}$$

Proof. We will give a proof for when n is even.

Let n be even. Since $j \in [1, n+1] \setminus 2\mathbb{Z}$, $n+1-j$ is even. So $\mathrm{inss}_j(b)$ is well-defined.

Now, since n is even, and j is odd, by the definition of the insertion-swapping function inss_j, $\mathrm{inss}_j(b)$ has odd numbers at odd positions and even numbers at even positions. Therefore,

$$\mathfrak{B}_{n+1} \supset \{\mathrm{inss}_j(b) \mid j \in [1, n+1] \setminus 2\mathbb{Z} \text{ and } b \in \mathfrak{B}_n\}.$$

On the other hand, let $u = u_1 \cdots u_{n+1} \in \mathfrak{B}_{n+1}$ with $u_i = n+1$. Then, i and $n+1$ are both odd. Then we have,

$$\overline{u} = \overline{u}_1 \cdots \overline{u}_n = u_1 \cdots u_{i-1} u_{i+2} u_{i+1} u_{i+4} u_{i+3} \cdots u_{n+1} u_n \in \mathfrak{B}_n.$$

Thus, $\mathrm{inss}_i(\overline{u}) = u$, which gives

$$\mathfrak{B}_{n+1} \subset \{\mathrm{inss}_j(b) \mid j \in [1, n+1] \setminus 2\mathbb{Z} \text{ and } b \in \mathfrak{B}_n\}.$$

For odd n, the proof is similar. □

Let $w \in \mathfrak{S}_n$, one can also see that the *inversion*

$$\mathrm{INV}(w) := |\{(w_i, w_j) \mid i < j \text{ and } w_i > w_j\}| \tag{7.6}$$

increases after applying ins_p and inss_q on w. Then, we have the following lemmas.

Lemma 7.6. *Let $w \in \mathfrak{S}_n$, and fix $p \in [1, n+1]$. Then, we have*

$$\mathrm{INV}(\mathrm{ins}_p(w)) = \mathrm{INV}(w) + n - p + 1.$$

Proof. Let $w = w_1 \cdots w_n$. Then $\mathrm{ins}_p w = w_1 \cdots w_{p-1}(n+1)w_p \cdots w_n$. Therefore, (w_i, w_j) counts an inversion in w if and only if (w_i, w_j) counts an inversion in $\mathrm{ins}_p(w)$. Since $n+1$ is at position p of $\mathrm{ins}_p(w)$, $(n+1, w_p), (n+1, w_{p+1}), \ldots, (n+1, w_n)$ count inversions in $\mathrm{ins}_p(w)$. Hence we have

$$\mathrm{INV}(\mathrm{ins}_p(w)) = \mathrm{INV}(w) + n - p + 1.$$

□

Lemma 7.7. *Let $w \in \mathfrak{S}_n$, and fix $q \in [1, n+1]$ such that $n-q+1$ is even. Then, we have*

$$\mathrm{INV}(\mathrm{inss}_q(w)) \geq \mathrm{INV}(w) + \frac{n-q+1}{2}.$$

Proof. By Lemma 7.6, inserting $n+1$ at position q creates $n-q+1$ inversions. After swapping the letters after position q, we lose at most $\frac{n-q+1}{2}$ inversions. Therefore,

$$\text{INV}(\text{inss}_q(w)) \geq \text{INV}(w) + n - q + 1 - \frac{n-q+1}{2} = \text{INV}(w) + \frac{n-q+1}{2}.$$

□

For example, consider $w = 126435$, and $p = q = 3$. Then, we have $n = 6$, and

$$\text{ins}_p(w) = 2176435, \quad \text{inss}_q(w) = 2174653.$$

We see that

$$\begin{aligned} \text{INV}(w) &= 4, \\ \text{INV}(\text{ins}_3(w)) &= 8 = \text{INV}(w) + n - p + 1, \\ \text{INV}(\text{inss}_3(w)) &= 8 \geq \text{INV}(w) + \frac{n-q+1}{2}. \end{aligned}$$

Observe that $|\mathfrak{A}_n| = |\tilde{\mathfrak{A}}_n| = \lfloor \frac{n}{2} \rfloor ! \lceil \frac{n}{2} \rceil !$. Then there exists a bijective map between $\mathfrak{A}_n$ and $\tilde{\mathfrak{A}}_n$. Let $w = w_1 \cdots w_n \in \mathfrak{S}_n$. Define two maps U and R given by

$$\begin{aligned} U : \mathfrak{S}_n &\to \mathfrak{S}_n, \\ w &\mapsto w^U := (n+1-w_1)(n+1-w_2)\cdots(n+1-w_n), \\ R : \mathfrak{S}_n &\to \mathfrak{S}_n, \\ w &\mapsto w^R := w_n \cdots w_1. \end{aligned}$$

We call the first map U the upside-down of w, and the second map R the reverse of w. For example if $w = 216435$, then $w^U = 561342$ and $w^R = 534612$. It is not difficult to see that the two maps are both involutions and the two maps commute. We abbreviate $U \circ R = R \circ U = RU$. This induces bijective maps between $\mathfrak{A}_n$ and $\tilde{\mathfrak{A}}_n$, and $\mathfrak{B}_n$ to itself.

Proposition 7.8. *The map $U \circ R : \mathfrak{S}_n \to \mathfrak{S}_n$ sending $w \mapsto w^{UR}$ restricts to a bijection between $\mathfrak{A}_n$, $\tilde{\mathfrak{A}}_n$, and to an involution on $\mathfrak{B}_n$.*

Proof. Consider the map $U \circ R\big|_{\mathfrak{A}_n}$, we claim that its inverse is $U \circ R\big|_{\tilde{\mathfrak{A}}_n}$.

To see this, consider $w \in \mathfrak{A}_n$, and let $t = \lfloor \frac{n}{2} \rfloor$. Since

$$\{w_{t+1}, \ldots, w_n\} = \{t+1, \ldots, n\},$$

we have

$$\{w_1^R, \ldots, w_t^R\} = \{t+1, \ldots, n\}.$$

Hence, we have

$$\{w_1^{RU}, \ldots, w_t^{RU}\} = \{1, \ldots, t\},.$$

Thus, we have $w^{RU} \in \tilde{\mathfrak{A}}_n$. Similarly, if $\tilde{w} \in \tilde{\mathfrak{A}}_n$, then $\tilde{w}^{RU} \in \mathfrak{A}_n$. Therefore, $U \circ R$ restricts to a bijection between $\mathfrak{A}_n$ and $\tilde{\mathfrak{A}}_n$.

Next, we show that $U \circ R\big|_{\mathfrak{B}_n}$ is an involution.

Let $u = u_1 \cdots u_n \in \mathfrak{B}_n$. Then, u_i and i have the same parity for all $1 \leq i \leq n$. Applying $U \circ R$ on u, we have

$$\tilde{u} := u^{RU} = (n+1-u_n)\cdots(n+1-u_1).$$

Since u_i and i have the same parity for all i,

$$n+1-u_i = 1+n-i+i-u_i \equiv 1+n-i \pmod 2. \tag{7.7}$$

Write $\tilde{u} = \tilde{u}_1 \cdots \tilde{u}_n$, then $\tilde{u}_j = n+1-u_{n+1-j}$ for all $1 \le j \le n$. By (7.7), we have

$$n+1-u_{n+1-j} \equiv 1+n-(n+1-j) = j \pmod 2.$$

Therefore, $\tilde{u}_j$ and j have the same parity. Therefore, $\tilde{u} \in \mathfrak{B}_n$. Now, apply $U \circ R\big|_{B-n}$ on $\tilde{u}$, we have

$$\tilde{u}^{RU} = (n+1-(n+1-u_1)) \cdots (n+1-(n+1-u_n)) = u_1 \cdots u_n = u.$$

Therefore $U \circ R$ restricts to an involution on $\mathfrak{B}_n$. □

In addition, the map $U \circ R$ preserves the Bruhat order.

Proposition 7.9. *Let $v,\ w \in \mathfrak{S}_n$ with $v \le w$ under the Bruhat order. Then, $v^{RU} \le w^{RU}$ under the Bruhat order.*

Proof. For all $u \in \mathfrak{S}_n$, $u^{RU} = w_0 v w_0$, where $w_0 = n(n-1)\cdots 1$. Also, we have $v \le w$ under the Bruhat order if and only if

$$w_0 v w_0 \le w_0 w w_0.$$

(See [1, p. 36–38].) □

One can also see that $U \circ R$ preserves inversions.

Lemma 7.10. *For all $w \in \mathfrak{S}_n$, $\textsc{inv}(w) = \textsc{inv}(w^{RU})$.*

Proof. Let $w \in \mathfrak{S}_n$ and $w^{RU} := v$. Then, we have

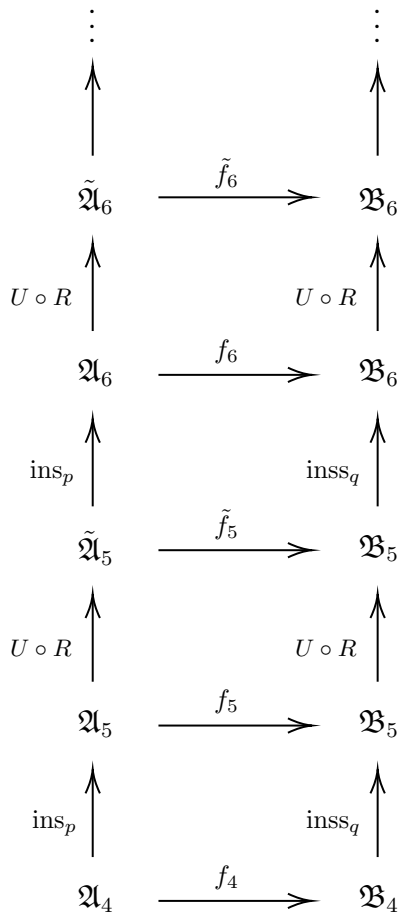
$$v = v_1 \cdots v_n = (n+1-w_n) \cdots (n+1-w_1).$$

Let (v_i, v_j) count as an inversion, then $(n+1-w_{n+1-i}, n+1-w_{n+1-j})$ counts as an inversion. Hence, $n+1-w_{n+1-i} > n+1-w_{n+1-j}$, which implies $w_{n+1-j} > w_{n+1-i}$. Since $i < j$, we have $n+1-j < n+1-i$. Then (w_{n+1-j}, w_{n+1-i}) counts as an inversion. Similarly, assume (w_i, w_j) is an inversion in w. Observe that $w = v^{RU}$, then (v_{n+1-j}, v_{n+1-i}) counts as an inversion in $v = w^{RU}$. Therefore, we have

$$\textsc{inv}(v) = \textsc{inv}(w^{RU}) = \textsc{inv}(w).$$

□

In order to show (7.2) is TNN, we aim to define a bijective map $f_n : \mathfrak{A}_n \to \mathfrak{B}_n$ such that $w \le f_n(w)$ for all $w \in \mathfrak{A}_n$. Although many such bijections appear in the literature, none is suitable for our purpose. (See [10].) Therefore, we define the required bijective maps inductively and illustrate them with a diagram.



To inductively define f_n and $\tilde{f}_n$, we start with the base case f_4 defined as follows.

$$1234 \longmapsto 1234 \quad 1243 \longmapsto 1432 \quad 2134 \longmapsto 3214 \quad 2143 \longmapsto 3412. \tag{7.8}$$

Before we define f_n with $n = 5, 6, \ldots$, we need an auxiliary map $\tilde{f}_n : \tilde{\mathfrak{A}}_n \to \mathfrak{B}_n$.

Let $\tilde{w} \in \tilde{\mathfrak{A}}_n$. By Proposition 7.8, there is a unique $w \in \mathfrak{A}_n$ such that $w^{RU} = \tilde{w}$. Since $|\mathfrak{A}_n| = |\mathfrak{B}_n| = |\tilde{\mathfrak{A}}_n| = \lfloor \frac{n}{2} \rfloor ! \lceil \frac{n}{2} \rceil !$, there exists a bijection between $\mathfrak{A}_n$ and $\mathfrak{B}_n$, and a bijection between $\tilde{\mathfrak{A}}_n$ and $\mathfrak{B}_n$. Denote $f_n : \mathfrak{A}_n \to \mathfrak{B}_n$ where when $n = 4$, f_4 is defined as above. Further, define $\tilde{f}_n$ by

$$\begin{aligned} \tilde{f}_n : \tilde{\mathfrak{A}}_n &\to \mathfrak{B}_n, \\ \tilde{w} &\mapsto [f_n(\tilde{w}^{RU})]^{RU}. \end{aligned} \tag{7.9}$$

It is not obvious that $\tilde{f}_n$ is surjective, which we now show.

Observation 7.11. *Given a bijection $f_n : \mathfrak{A}_n \to \mathfrak{B}_n$, the map $\tilde{f}_n : \tilde{\mathfrak{A}}_n \to \mathfrak{B}_n$ is a bijection.*

Proof. By Proposition 7.8, if $b \in \mathfrak{B}_n$, then $b^{RU} \in \mathfrak{B}_n$. Therefore, the inverse of $\tilde{f}_n$ can be given by

$$\begin{aligned} \tilde{f}_n^{-1} : \mathfrak{B}_n &\to \tilde{\mathfrak{A}}_n, \\ b &\mapsto [f_n^{-1}(b^{RU})]^{RU}. \end{aligned}$$

One can check that $\tilde{f}_n \circ \tilde{f}_n^{-1} = \tilde{f}_n^{-1} \circ \tilde{f}_n = \mathrm{id}$. □

By our construction of $\tilde{f}_n$, many properties of f_n are inherited in $\tilde{f}_n$.

Lemma 7.12. *Let $f_n : \mathfrak{A}_n \to \mathfrak{B}_n$ be a bijection such that $w \leq f_n(w)$ for all $w \in \mathfrak{A}_n$. Then, the bijection $\tilde{f}_n$ (7.9) satisfies $\tilde{w} \leq \tilde{f}_n(\tilde{w})$ for every $\tilde{w} \in \tilde{\mathfrak{A}}_n$.*

Proof. Let $w \in \mathfrak{A}_n$ with $w^{RU} = \tilde{w}$. By Observation 7.11, $\tilde{f}_n$ is a bijection. By Proposition 7.9, $\tilde{w} = w^{RU} \leq f_n(w)^{RU} = \tilde{f}_n(\tilde{w})$. □

In addition, if f_n increases the number of inversions, so does $\tilde{f}_n$.

Lemma 7.13. *Let $f_n : \mathfrak{A}_n \to \mathfrak{B}_n$ be a bijection such that $\textsc{inv}(w) \leq \textsc{inv}(f_n(w))$ for all $w \in \mathfrak{A}_n$. Then, the bijection $\tilde{f}_n$ satisfies $\textsc{inv}(\tilde{w}) \leq \textsc{inv}(\tilde{f}_n(\tilde{w}))$ for every $\tilde{w} \in \tilde{\mathfrak{A}}_n$.*

Proof. Let $\tilde{w} \in \tilde{\mathfrak{A}}_n$ be arbitrary. Then, by Proposition 7.8, $\tilde{w}^{RU} := w \in \mathfrak{A}_n$. By Lemma 7.10,

$$\textsc{inv}(\tilde{w}) = \textsc{inv}(w) \text{ and } \textsc{inv}(f_n(w)) = \textsc{inv}([f_n(w)]^{RU}).$$

Since $\textsc{inv}(w) \leq \textsc{inv}(f_n(w))$ for all $w \in \mathfrak{A}_n$, we have

$$\textsc{inv}(\tilde{w}) \leq \textsc{inv}([f_n(w)]^{RU}) = \textsc{inv}([f_n(\tilde{w}^{RU})]^{RU}).$$

By (7.9), $[f_n(\tilde{w}^{RU})]^{RU} = \tilde{f}(\tilde{w})$. Hence, $\textsc{inv}(\tilde{w}) \leq \textsc{inv}(\tilde{f}(\tilde{w}))$. □

One can see from the proof that Lemma 7.13 remains correct if we replace all weak inequalities by strict inequalities and restrict f_n on non-identity elements of $\mathfrak{A}_n$, i.e, $\mathfrak{A}_n \setminus \{e\}$.

Corollary 7.14. *Let $f_n : \mathfrak{A}_n \to \mathfrak{B}_n$ be a bijection such that $\textsc{inv}(w) < \textsc{inv}(f_n(w))$ for all $w \in \mathfrak{A}_n \setminus \{e\}$. Then, the bijection $\tilde{f}_n$ satisfies $\textsc{inv}(\tilde{w}) < \textsc{inv}(\tilde{f}_n(\tilde{w}))$ for every $\tilde{w} \in \tilde{\mathfrak{A}}_n \setminus \{e\}$.*

When $n \leq 3$, it is easy to find a bijection $f_n : \mathfrak{A}_n \to \mathfrak{B}_n$ such that $w \leq f_n(w)$. Also, one can easily test if a given polynomial is in the 3×3 immanant space is TNN [20]. With our base case set to $n = 4$, we are now ready to introduce the following recursive algorithm to define f_n for $n = 5, 6, \ldots$.

Algorithm 7.15. Let $\mathfrak{A}_n$, $\tilde{\mathfrak{A}}_n$, $\mathfrak{B}_n$ be defined as above.

Base Case: Let f_4 be the map

$$1234 \longmapsto 1234 \quad 1243 \longmapsto 1432 \quad 2134 \longmapsto 3214 \quad 2143 \longmapsto 3412. \tag{7.10}$$

For $n = 5, 6, 7, \ldots$ do the following.

(1) If n is odd,
   a. Write $n - 1 = 2k$.
   b. For each $w \in \mathfrak{A}_n$, find the unique pair (by Observation 7.3) $(a, p) \in \mathfrak{A}_{n-1} \times \mathbb{N}$ such that $w = \mathrm{ins}_p(a)$.
   c. Define

$$f_n(w) = \mathrm{inss}_{2(p-k)-1}(f_{n-1}(a)). \tag{7.11}$$

(2) If n is even,
   a. Write $n - 1 = 2k + 1$.
   b. Define $\tilde{f}_{n-1} : \tilde{\mathfrak{A}}_n \to \mathfrak{B}_n$ by f_{n-1} as in (7.9).
   c. For each $u \in \mathfrak{A}_n$, find the unique pair (by Observation 7.4) $(\tilde{w}, q) \in \tilde{\mathfrak{A}}_n \times \mathbb{N}$ such that $u = \mathrm{ins}_q(\tilde{w})$.

d. Define

$$f_n(u) = \mathrm{inss}_{2(q-k-1)}(\tilde{f}_{n-1}(\tilde{w})). \tag{7.12}$$

As an example, let us look at the function f_5 defined by Algorithm 7.15. Now, we can define f_5 from the base case (7.10). In this case, we have $n = 5$ and $k = 2$. For example, to find the image of $w = 21534 \in \mathfrak{A}_5$ under f_5, we have the unique pair $(a, p) = (2134, 3)$, i.e., $21534 = \mathrm{ins}_3(2134)$. By (7.11), we have

$$f_5(21534) = \mathrm{inss}_{2(3-2)-1}(f_4(2134)) = \mathrm{inss}_1(3214) = 52341.$$

Repeat the process for every element in $\mathfrak{A}_5$, and we get the map f_5 as follows.

$12534 \longmapsto 52143$	$12354 \longmapsto 12543$	$12345 \longmapsto 12345$
$12543 \longmapsto 54123$	$12453 \longmapsto 14523$	$12435 \longmapsto 14325$
$21534 \longmapsto 52341$	$21354 \longmapsto 32541$	$21345 \longmapsto 32145$
$21543 \longmapsto 54321$	$21453 \longmapsto 34521$	$21435 \longmapsto 34125.$

To define f_6, we need to find $\tilde{f}_5$ from f_5. By (7.9), we get $\tilde{f}_5$ as follows.

$23145 \longmapsto 32541$	$21345 \longmapsto 32145$	$12345 \longmapsto 12345$
$32145 \longmapsto 34521$	$31245 \longmapsto 34125$	$13245 \longmapsto 14325$
$23154 \longmapsto 52341$	$21354 \longmapsto 52143$	$12354 \longmapsto 12543$
$32154 \longmapsto 54321$	$31254 \longmapsto 54123$	$13254 \longmapsto 14523.$

Now, we can define f_6 from $\tilde{f}_5$. In this case, $n = 6$ and $k = 2$. For example, to find the image of $u = 231564 \in \mathfrak{A}_6$ under f_6, we have the unique pair $(\tilde{w}, q) = (23154, 5)$, i.e., $231564 = \mathrm{ins}_5(23154)$. From the above definition of $\tilde{f}_5$, we have $\tilde{f}_5(23154) = 52341$. Then, by (7.12), we have

$$f_6(231564) = \mathrm{inss}_{2(5-2-1)}(\tilde{f}_5(23154)) = \mathrm{inss}_4(52341) = 523614.$$

Repeat the process for every element in $\mathfrak{A}_6$, and we get the map f_6 as follows.

$231645 \longmapsto 365214$	$231465 \longmapsto 325614$	$231456 \longmapsto 325416$
$321645 \longmapsto 365412$	$321465 \longmapsto 345612$	$321456 \longmapsto 345216$
$231654 \longmapsto 563214$	$231564 \longmapsto 523614$	$231546 \longmapsto 523416$
$321654 \longmapsto 563412$	$321564 \longmapsto 543612$	$321546 \longmapsto 543216$
$213645 \longmapsto 361254$	$213465 \longmapsto 321654$	$213456 \longmapsto 321456$
$312645 \longmapsto 361452$	$312465 \longmapsto 341652$	$312456 \longmapsto 341256$
$213654 \longmapsto 561234$	$213564 \longmapsto 521634$	$213546 \longmapsto 521436$
$312654 \longmapsto 561432$	$312564 \longmapsto 541632$	$312546 \longmapsto 541236$
$123645 \longmapsto 163254$	$123465 \longmapsto 123654$	$123456 \longmapsto 123456$
$132645 \longmapsto 163452$	$132465 \longmapsto 143652$	$132456 \longmapsto 143256$
$123654 \longmapsto 165234$	$123564 \longmapsto 125634$	$123546 \longmapsto 125436$
$132654 \longmapsto 165432$	$132564 \longmapsto 145632$	$132546 \longmapsto 145236.$

To show Algorithm 7.15 is well-defined, it suffices to show that f_n is bijective for $n = 5, 6, 7, \ldots$.

Lemma 7.16. *In Algorithm 7.15, if $n-1 = 2k$ is even and f_{n-1} is a bijection, then f_n is a bijection.*

Proof. Suppose $n-1 = 2k$ is even and let f_{n-1} be a bijection in Algorithm 7.15. Let $u \in \mathfrak{B}_n$. Then, by Observation 7.5, there is a unique pair $(b, q) \in \mathfrak{B}_{n-1} \times \mathbb{N}$ such that $u = \mathrm{inss}_q(b)$. Since f_{n-1} is a bijection, we can find $f_{n-1}^{-1}(b) \in \mathfrak{A}_{n-1}$. We define $g_n : \mathfrak{B}_n \to \mathfrak{A}_n$ by

$$g_n(u) = \mathrm{ins}_{k+\frac{q+1}{2}}(f_{n-1}^{-1}(b)). \tag{7.13}$$

Next, we claim that g_n is the inverse function of f_n.

To see this, let $w \in \mathfrak{A}_n$ be arbitrary. By (7.11),

$$f_n(w) = \mathrm{inss}_{2(p-k)-1}(f_{n-1}(a)),$$

where $(a, p) \in \mathfrak{A}_{n-1} \times \mathbb{N}$ with $\mathrm{ins}_p(a) = w$. To find $g_n(f_n(w))$, we need to find the unique pair (b, q) in (7.13). We see that $(b, q) = (f_{n-1}(a), 2(p-k)-1)$. Hence, we have

$$g_n(f_n(w)) = \mathrm{ins}_{k+\frac{2(p-k)-1+1}{2}}(f_{n-1}^{-1}(f_{n-1}(a))) = \mathrm{ins}_p(a) = w.$$

On the other hand, let $u \in \mathfrak{B}_n$ be arbitrary, then

$$g_n(u) = \mathrm{ins}_{k+\frac{q+1}{2}}(f_{n-1}^{-1}(b)),$$

where $(b, q) \in \mathfrak{B}_{n-1} \times \mathbb{N}$ with $u = \mathrm{inss}_q(b)$. To find $f_n(g_n(u))$, we need to find the unique pair (a, p) in (7.11). We see that $(a, p) = (f_{n-1}^{-1}(b), k + \frac{q+1}{2})$. Hence by (7.11), we have

$$f_n(g_n(u)) = \mathrm{inss}_{2(k+\frac{q+1}{2}-k)-1}(f_{n-1}(f_{n-1}^{-1}(b))) = \mathrm{inss}_q(b) = u.$$

Therefore, g_n is the inverse function of f_n. □

Continuing the example from earlier, now to find the image of $u = 52341 \in \mathfrak{B}_5$ under f_5^{-1}, we have the unique pair $(b, q) = (3214, 1)$, i.e., $52341 = \mathrm{inss}_1(3214)$. By (7.13), we have

$$f_5^{-1}(52341) = \mathrm{ins}_{2+\frac{1+1}{2}} f_4^{-1}(3214) = \mathrm{ins}_3(2134) = 21534.$$

Lemma 7.17. *In Algorithm 7.15, if $n-1 = 2k+1$ is odd and f_{n-1} is a bijection, then f_n is a bijection.*

Proof. Suppose $n-1 = 2k+1$ is odd and let f_{n-1} be a bijection in Algorithm 7.15. By Observation 7.11, $\tilde{f}_{n-1} : \tilde{\mathfrak{A}}_{n-1} \to \mathfrak{B}_{n-1}$ is a bijection. Let $u \in \mathfrak{B}_n$. Then, by Observation 7.5, there is a unique pair $(b, p) \in \mathfrak{B}_{n-1} \times \mathbb{N}$ such that $u = \mathrm{inss}_p(b)$. Since $\tilde{f}_{n-1}$ is a bijection, we can find $\tilde{f}_{n-1}^{-1}(b) \in \tilde{\mathfrak{A}}_{n-1}$. We define $g_n : \mathfrak{B}_n \to \mathfrak{A}_n$ by

$$g_n(u) = \mathrm{ins}_{k+1+\frac{p}{2}}(\tilde{f}_{n-1}^{-1}(b)). \tag{7.14}$$

Next, we claim that g_n is the inverse function of f_n.

To see this, let $w \in \mathfrak{A}_n$ be arbitrary. By (7.12),

$$f_n(w) = \mathrm{inss}_{2(q-k-1)}(\tilde{f}_{n-1}(\tilde{w})),$$

where $(\tilde{w}, q) \in \tilde{\mathfrak{A}}_n \times \mathbb{N}$ with $w = \mathrm{ins}_q(\tilde{w})$. To find $g_n(f_n(w))$, we need to find the unique pair (b, p) in (7.14). We see that $(b, p) = (\tilde{f}_{n-1}(\tilde{w}), 2(q-k-1))$. Hence by (7.14), we have

$$g_n(f_n(w)) = \mathrm{ins}_{k+1+\frac{2(q-k-1)}{2}}(\tilde{f}_{n-1}^{-1}(\tilde{f}_{n-1}(\tilde{w}))) = \mathrm{ins}_q(\tilde{w}) = w.$$

On the other hand, let $u \in \mathfrak{B}_n$ be arbitrary, then

$$g_n(u) = \mathrm{ins}_{k+1+\frac{p}{2}}(\tilde{f}_{n-1}^{-1}(b)),$$

where $(b,p) \in \mathfrak{B}_{n-1} \times \mathbb{N}$ with $u = \mathrm{inss}_p(b)$. To find $f_n(g_n(u))$, we need to find the unique pair $(\tilde{w}, q)$ in (7.12). We see that $(\tilde{w}, q) = (\tilde{f}_{n-1}^{-1}(b), k+1+\frac{p}{2})$. Hence, by (7.12), we have

$$f_n(g_n(u)) = \mathrm{inss}_{2(k+1+\frac{p}{2}-k-1)}(\tilde{f}_{n-1}(\tilde{f}_{n-1}^{-1}(b))) = \mathrm{inss}_p(b) = u.$$

Therefore, g_n is the inverse function of f_n. □

For example, to find the image of $u = 523614 \in \mathfrak{B}_6$ under f_6^{-1}, we have the unique pair $(b,p) = (52341, 4)$, i.e., $523614 = \mathrm{inss}_4(52341)$. By (7.14), we have

$$f_6^{-1}(523614) = \mathrm{ins}_{2+1+\frac{4}{2}}(\tilde{f}_5^{-1}(52341)) = \mathrm{ins}_5(23154) = 231564.$$

Then, we have the immediate corollary.

Corollary 7.18. *Algorithm 7.15 is well-defined.*

Proof. As the base case, one can observe that f_4 is a bijection. Then inductively from Lemma 7.16 and Lemma 7.17, we have the desired result. □

We aim to build a bijection $\mathfrak{A}_n \to \mathfrak{B}_n$ such that $w \leq f_n(w)$ for all $w \in \mathfrak{A}_n$. It's easy to see that the identity element $e = 123\cdots n \in \mathfrak{S}_n$ is in both $\mathfrak{A}_n$ and $\mathfrak{B}_n$. Therefore, the identity element has to be sent to the identity under f_n, and we have the next observation.

Observation 7.19. *The bijection f_n defined in Algorithm 7.15 sends identity to identity.*

Proof. When $n = 4$, we see that $1234 \longmapsto 1234$. We assume that f_{n-1} sends identity to identity, and consider f_n. Let $e_n := 123\cdots n$ be the identity element in $\mathfrak{S}_n$. If n is odd, then $e_n = \mathrm{ins}_n(e_{n-1})$. Write $n-1 = 2k$, by (7.11), we have

$$f_n(e_n) = \mathrm{inss}_{2(n-k)-1}(f_{n-1}(e_{n-1})) = \mathrm{inss}_n(e_{n-1}) = e_n.$$

If n is even, then $e_n = \mathrm{ins}_n(e_{n-1})$. Since f_{n-1} sends identity to identity, by (7.9) the bijection $\tilde{f}_{n-1}$ sends identity to identity. Write $n-1 = 2k+1$, by (7.12), we have

$$f_n(e_n) = \mathrm{inss}_{2(n-k-1)}(\tilde{f}_{n-1}(e_{n-1})) = \mathrm{inss}_n(e_{n-1}) = e_n. \tag{7.15}$$

Therefore, f_n sends identity to identity. □

By the definition of $\tilde{f}_n$ in (7.9), we have the immediate corollary.

Corollary 7.20. *The bijection $\tilde{f}_n$, where f_n is defined in Algorithm 7.15, sends identity to identity.*

It is well-known that the Bruhat lattice is graded by the number of inversions (7.6). We have seen that f_n sends the identity element to the identity element. For $w \in \mathfrak{A}_n \setminus \{e\}$, $f_n(w)$ is located higher than w in the Bruhat lattice of $\mathfrak{S}_n$.

Theorem 7.21. *In Algorithm 7.15, for all $w \in \mathfrak{A}_n \setminus \{e\}$, we have $\mathrm{INV}(w) < \mathrm{INV}(f_n(w))$.*

Proof. Let $w \in \mathfrak{A}_n$ not equal to identity. One can check that f_4 satisfies $\mathrm{INV}(w) < \mathrm{INV}(f_4(w))$ for all non-identity elements in $\mathfrak{A}_4$.

Now, we are going to use induction. We assume that $\mathrm{INV}(w) < \mathrm{INV}(f_{n-1}(w))$ for all non-identity $v \in \mathfrak{A}_{n-1}$ as our induction hypothesis, and consider the map f_n.

Case 1: Fix $n-1 = 2k$ and let $v \in \mathfrak{A}_n$ not equal to identity. There exists a unique pair $(a,p) \in \mathfrak{A}_{n-1} \times \mathbb{N}$ such that $v = \mathrm{ins}_p(a)$. By (7.11), $f_n(v) = \mathrm{inss}_{2(p-k)-1}(f_{n-1}(a))$. By Lemma 7.7, we have $\mathrm{INV}(f_n(v)) = \mathrm{INV}(\mathrm{inss}_{2(p-k)-1}(f_{n-1}(a)))$ is at least

$$\mathrm{INV}(f_{n-1}(a)) + \frac{n-1-(2(p-k-1)-1)+1}{2} = \mathrm{INV}(f_{n-1}(a)) + n - p,$$

with strict inequality when $f_{n-1}(a)$ equals identity, i.e., a equals identity by Observation 7.19. If a is not identity, by our induction hypothesis, we have

$$\mathrm{INV}(f_n(v)) \geq \mathrm{INV}(f_{n-1}(a)) + n - p > \mathrm{INV}(a) + n - 1 - p + 1.$$

If a is identity, we have

$$\mathrm{INV}(f_n(v)) > \mathrm{INV}(f_{n-1}(a)) + n - p = \mathrm{INV}(a) + n - 1 - p + 1.$$

By Lemma 7.6, we see that

$$\mathrm{INV}(a) + n - 1 - p + 1 = \mathrm{INV}(\mathrm{ins}_p(a)) = \mathrm{INV}(v).$$

Therefore, $\mathrm{INV}(f_n(v)) > \mathrm{INV}(v)$.

Case 2: Fix $n-1 = 2k+1$ and let $u \in \mathfrak{A}_n$ not equal to identity. There exists a unique pair $(\tilde{u}, q) \in \tilde{\mathfrak{A}}_{n-1} \times \mathbb{N}$ such that $u = \mathrm{ins}_q(\tilde{u})$. By (7.12), $f_n(u) = \mathrm{inss}_{2(q-k-1)}(\tilde{f}_{n-1}(\tilde{u}))$. By Lemma 7.7, $\mathrm{INV}(f_n(u))$ is at least

$$\mathrm{INV}(\tilde{f}_{n-1}(\tilde{u})) + \frac{n-1-2(q-k-1)+1}{2} = \mathrm{INV}(\tilde{f}_{n-1}(\tilde{u})) + n - q,$$

with strict inequality when $\tilde{f}_{n-1}(\tilde{u})$ equals identity, i.e.,$\tilde{u}$ equals identity by Corollary 7.20. If $\tilde{u}$ is not identity, by Corollary 7.14 and our induction hypothesis, we have

$$\mathrm{INV}(\tilde{f}_{n-1}(\tilde{u})) > \mathrm{INV}(\tilde{u}).$$

Hence, we have

$$\mathrm{INV}(f_n(u)) \geq \mathrm{INV}(\tilde{f}_{n-1}(\tilde{u})) + n - q > \mathrm{INV}(\tilde{u}) + n - q.$$

If $\tilde{u}$ is identity, we have

$$\mathrm{INV}(f_n(u)) > \mathrm{INV}(\tilde{f}_{n-1}(\tilde{u})) + n - q = \mathrm{INV}(\tilde{u}) + n - 1 - q + 1.$$

By Lemma 7.6, we have

$$\mathrm{INV}(\tilde{u}) + n - 1 - q + 1 = \mathrm{INV}(\mathrm{ins}_q(\tilde{u})) = \mathrm{INV}(u).$$

Therefore, $\mathrm{INV}(f_n(u)) > \mathrm{INV}(u)$. □

We know that $\mathrm{INV}(v) < \mathrm{INV}(w)$ is a necessary condition of $v < w$ in the Bruhat order. Hence,we have the immediate corollary.

Corollary 7.22. *In Algorithm 7.15, for all $w \in \mathfrak{A}_n \setminus \{e\}$, we have $w \not\geq f_n(w)$ under the Bruhat order.*

The next theorem is due to Ehresmann [6], which is a way to compare two elements in $\mathfrak{S}_n$ under the Bruhat order. It is also called the tableau criterion.

Theorem 7.23 (Tableau Criterion)**.** *Let $w, u \in \mathfrak{S}_n$. Define $w_{i,k}$ to be the i-th element in the increasing rearrangement of $w_1, w_2, \ldots, w_k$ and similarly define $u_{i,k}$. Then $w \leq u$ under the Bruhat order if and only if $w_{i,k} \leq u_{i,k}$ as an integer for all $k \in [1, n]$ and $1 \leq i \leq k$.*

For example, we can use Theorem 7.23 to compare 2143 and 3241 in the Bruhat order. When $k = 1, 2, 3, 4$, we have the following rearrangements of 2143 and 3241.

	2143	3241
$k = 1$	2	2
$k = 2$	1, 2	2, 3
$k = 3$	1, 2, 4	2, 3, 4
$k = 4$	1, 2, 3, 4	1, 2, 3, 4

One can see that for each k, the i-th letter of the rearrangement of 2143 is less than or equal to the i-th letter of the rearrangement of 3241. Therefore, $2143 \leq 3241$ in the Bruhat order.

By the Tableau Criterion, we can compare every $w \in \mathfrak{A}_n$ and its image $f_n(w) \in \mathfrak{B}_n$ under Algorithm 7.15. Then, we have the following theorem.

Theorem 7.24. *In Algorithm 7.15, for $n \leq 13$, we have $w \leq f_n(w)$ under the Bruhat order for all $w \in \mathfrak{A}_n$.*

Proof. This is done by brute force, with MATLAB code in the appendix. □

We are going to extend Theorem 7.24 to all n in the next section.

8. Bruhat Order Comparison for the Bijection f_n

In this section, we are going to prove that $w \leq f_n(w)$ under the Bruhat order for all $w \in \mathfrak{A}_n$, where f_n is defined in Algorithm 7.15.

Now, we consider the case when $n = 2k$. In this case, both families of permutations under consideration can be naturally parametrized by pairs of permutations in $\mathfrak{S}_k$. More precisely, an element of the Young subgroup preserving the two blocks

$$[1, k] \qquad \text{and} \qquad [k+1, \ldots, 2k]$$

is determined by the order of the entries within each block. Thus, for $\sigma, \tau \in \mathfrak{S}_k$, we write

$$A_k(\sigma, \tau) = \sigma_1 \cdots \sigma_k \, (k + \tau_1) \cdots (k + \tau_k) \in \mathfrak{A}_{2k}. \tag{8.1}$$

Similarly, a parity-alternating permutation is determined by the relative order of its odd entries and its even entries. Accordingly, for $\alpha, \beta \in \mathfrak{S}_k$, we write

$$B_k(\alpha, \beta) = (2\alpha_1 - 1)(2\beta_1) \cdots (2\alpha_k - 1)(2\beta_k) \in \mathfrak{B}_{2k}. \tag{8.2}$$

These parametrizations will allow us to describe the map between the two families in terms of permutations in $\mathfrak{S}_k$.

The parametrizations A_k and B_k also behave naturally under the RU-operation. In both cases, applying RU reverses the roles of the two component permutations and applies RU to each of them. We record these identities in the following lemma.

Lemma 8.1. *Let $\sigma, \tau, \alpha, \beta \in \mathfrak{S}_k$. Then we have*

$$A_k(\sigma, \tau)^{RU} = A_k(\tau^{RU}, \sigma^{RU}), \tag{8.3}$$

and

$$B_k(\alpha, \beta)^{RU} = B_k(\beta^{RU}, \alpha^{RU}). \tag{8.4}$$

Proof. By definition,

$$A_k(\sigma,\tau) = \sigma_1 \cdots \sigma_k (k+\tau_1) \cdots (k+\tau_k).$$

Applying RU, we first reverse the order of the entries and then replace each entry x by $2k+1-x$. Thus,

$$A_k(\sigma,\tau)^{RU} = (k+1-\tau_k) \cdots (k+1-\tau_1)(2k+1-\sigma_k) \cdots (2k+1-\sigma_1).$$

Since $\tau^{RU} = (k+1-\tau_k) \cdots (k+1-\tau_1)$ and $\sigma^{RU} = (k+1-\sigma_k) \cdots (k+1-\sigma_1)$, we have

$$2k+1-\sigma_i = k + (k+1-\sigma_i).$$

Therefore,

$$A_k(\sigma,\tau)^{RU} = A_k(\tau^{RU}, \sigma^{RU}),$$

which proves (8.3).

Similarly, $B_k(\alpha,\beta) = (2\alpha_1 - 1)(2\beta_1) \cdots (2\alpha_k - 1)(2\beta_k)$. Applying RU gives

$$B_k(\alpha,\beta)^{RU} = \big(2(k+1-\beta_k)-1\big)\big(2(k+1-\alpha_k)\big) \cdots \big(2(k+1-\beta_1)-1\big)\big(2(k+1-\alpha_1)\big).$$

Since $\beta^{RU} = (k+1-\beta_k) \cdots (k+1-\beta_1)$ and $\alpha^{RU} = (k+1-\alpha_k) \cdots (k+1-\alpha_1)$, we have

$$B_k(\alpha,\beta)^{RU} = B_k(\beta^{RU}, \alpha^{RU}).$$

□

For example, let $k = 3$, $\sigma = 123$, and $\tau = 231$. Then

$$A_3(123, 231) = 123564.$$

Applying RU, we obtain

$$123564^{RU} = 312456.$$

On the other hand,

$$\tau^{RU} = 312, \qquad \sigma^{RU} = 123,$$

and hence

$$A_3(\tau^{RU}, \sigma^{RU}) = A_3(312, 123) = 312456.$$

Similarly, let $\alpha = 231$ and $\beta = 132$. Then

$$B_3(231, 132) = 325614,$$

and

$$325614^{RU} = 361254.$$

Since

$$\beta^{RU} = 213, \qquad \alpha^{RU} = 312,$$

we also have

$$B_3(\beta^{RU}, \alpha^{RU}) = B_3(213, 312) = 361254.$$

Notice that in both identities above, the two component permutations are interchanged after applying RU.

For $d \in [k+1]$, define a minimum-insertion map

$$I_d^- : \mathfrak{S}_k \to \mathfrak{S}_{k+1},$$
$$w = w_1 \cdots w_k \mapsto (w_1+1) \cdots (w_{d-1}+1)\, 1\, (w_d+1) \cdots (w_k+1).$$

Thus, I_d^- inserts the new minimum 1 in position d, after increasing each original entry by 1.

For convenience, we also write

$$I_h^+ := \mathrm{ins}_h,$$

where ins_h is the insertion map defined in Section 7.

The insertion maps interact with the RU-operation in a simple way. We record the following two identities for later use.

Lemma 8.2. *For $w \in \mathfrak{S}_k$, and $d, h \in [k+1]$, we have*

$$\big(I_d^-(w)\big)^{RU} = I_{k+2-d}^+\big(w^{RU}\big), \text{ and } \big(I_h^+(w)\big)^{RU} = I_{k+2-h}^-\big(w^{RU}\big).$$

Proof. The first identity follows directly from the definition of RU: reversing the positions sends the entry inserted in position d to position $k+2-d$, while complementing the entries sends the new minimum 1 to the new maximum $k+1$. The proof of the second identity is similar. □

Using the parametrizations A_k and B_k above, we may equivalently encode the map f_{2k} defined in Algorithm 7.15 as a map

$$F_k : S_k \times S_k \longrightarrow S_k \times S_k.$$

More precisely, for $\sigma, \tau, \alpha, \beta \in S_k$, we define

$$F_k(\sigma, \tau) = (\alpha, \beta) \quad \Longleftrightarrow \quad f_{2k}\big(A_k(\sigma, \tau)\big) = B_k(\alpha, \beta). \tag{8.5}$$

Thus, F_k records the action of f_{2k} in terms of the two S_k-coordinates determined by the parametrizations A_k and B_k.

For example, let $k = 3$ and take $\sigma = \tau = 231$. Then

$$A_3(231, 231) = 231564.$$

From the definition of f_6, we have

$$f_6(231564) = 523614.$$

On the other hand,

$$523614 = B_3(321, 132).$$

Therefore,

$$F_3(231, 231) = (321, 132).$$

Indeed, the odd entries of 523614 occur in the order $5, 3, 1$, corresponding to 321, while the even entries occur in the order $2, 6, 4$, corresponding to 132.

Although our main focus is the case $n = 2k$, the recursive definition of f_{2k} naturally involves permutations in $\mathfrak{S}_{2k-1}$. We therefore introduce a corresponding parametrization for parity-alternating permutations of odd size. For $\alpha \in \mathfrak{S}_k$ and $\beta \in \mathfrak{S}_{k-1}$, define

$$B_k^\circ(\alpha, \beta) = (2\alpha_1 - 1)(2\beta_1) \cdots (2\alpha_{k-1} - 1)(2\beta_{k-1})(2\alpha_k - 1). \tag{8.6}$$

Thus, $B_k^\circ(\alpha, \beta) \in \mathfrak{B}_{2k-1}$ is the parity-alternating permutation whose odd entries are ordered according to α and whose even entries are ordered according to β.

There are some properties involving B_k° and maps we defined earlier.

Lemma 8.3. *The insertion-swapping maps satisfy the following identities.*

(i) Let $w, v \in \mathfrak{S}_{k-1}$ and $s \in [k]$. Then

$$\mathrm{inss}_{2s-1}\big(B_{k-1}(w, v)\big) = B_k^\circ\big(I_s^+(w), v\big).$$

(ii) Let $w \in \mathfrak{S}_k$, $v \in \mathfrak{S}_{k-1}$, and $h \in [k]$. Then

$$\mathrm{inss}_{2h}\big(B_k^\circ(w,v)\big) = B_k\big(w, I_h^+(v)\big).$$

Proof. For part (i), we have

$$B_{k-1}(w,v) = (2w_1-1)(2v_1)\cdots(2w_{s-1}-1)(2v_{s-1})(2w_s-1)(2v_s)\cdots(2w_{k-1}-1)(2v_{k-1}).$$

By the definition of inss_{2s-1},

$$\begin{aligned}\mathrm{inss}_{2s-1}\big(B_{k-1}(w,v)\big) = (2w_1-1)(2v_1)\cdots(2w_{s-1}-1)(2v_{s-1})(2k-1)\\ (2v_s)(2w_s-1)(2v_{s+1})(2w_{s+1}-1)\cdots(2v_{k-1})(2w_{k-1}-1).\end{aligned}$$

Since

$$I_s^+(w) = w_1\cdots w_{s-1}\,k\,w_s\cdots w_{k-1},$$

the right-hand side is precisely

$$B_k^\circ\big(I_s^+(w), v\big).$$

For part (ii), we have

$$B_k^\circ(w,v) = (2w_1-1)(2v_1)\cdots(2w_{h-1}-1)(2v_{h-1})(2w_h-1)(2v_h)\cdots(2w_k-1).$$

Applying inss_{2h} gives

$$\begin{aligned}\mathrm{inss}_{2h}\big(B_k^\circ(w,v)\big) = (2w_1-1)(2v_1)\cdots(2w_{h-1}-1)(2v_{h-1})(2w_h-1)(2k)\\ (2w_{h+1}-1)(2v_h)\cdots(2w_k-1)(2v_{k-1}).\end{aligned}$$

Since

$$I_h^+(v) = v_1\cdots v_{h-1}\,k\,v_h\cdots v_{k-1},$$

this is exactly

$$B_k\big(w, I_h^+(v)\big).$$

□

For example, let $k=3$, $w=21$, and $v=12$. For part (i), take $s=1$. Then

$$B_2(21,12) = 3214$$

and

$$I_1^+(21) = 321.$$

Hence

$$\mathrm{inss}_1\big(B_2(21,12)\big) = \mathrm{inss}_1(3214) = 52341 = B_3^\circ(321,12).$$

For part (ii), take $w=321$, $v=12$, and $h=2$. We have

$$B_3^\circ(321,12) = 52341$$

and

$$I_2^+(12) = 132.$$

Therefore,

$$\mathrm{inss}_4\big(B_3^\circ(321,12)\big) = \mathrm{inss}_4(52341) = 523614 = B_3(321,132).$$

Lemma 8.4. *Let $w \in \mathfrak{S}_k$ and $v \in \mathfrak{S}_{k-1}$. Then*

$$\big(B_k^\circ(w,v)\big)^{RU} = B_k^\circ\big(w^{RU}, v^{RU}\big).$$

Proof. By definition, we have

$$\begin{aligned} B_k^\circ(w,v) &= (2w_1-1)(2v_1)\cdots(2w_{k-1}-1)(2v_{k-1})(2w_k-1),\\ w^{RU} &= (k+1-w_k)\cdots(k+1-w_1),\\ v^{RU} &= (k-v_{k-1})\cdots(k-v_1). \end{aligned}$$

Applying RU on $\big(B_k^\circ(w,v)\big)$, we have

$$\begin{aligned} \big(B_k^\circ(w,v)\big)^{RU} = \big(2(k+1-w_k)-1\big)\big(2(k-v_{k-1})\big)\cdots\\ \big(2(k+1-w_2)-1\big)\big(2(k-v_1)\big)\big(2(k+1-w_1)-1\big), \end{aligned}$$

which equals $B_k^\circ\big(w^{RU},v^{RU}\big)$ by definition of B_k°. □

For example, let $k=3$, $w=231$, and $v=21$. Then

$$B_3^\circ(231,21)=34521.$$

Applying RU gives

$$34521^{RU}=54123.$$

On the other hand,

$$231^{RU}=312,\qquad 21^{RU}=21,$$

and hence

$$B_3^\circ(231^{RU},21^{RU})=B_3^\circ(312,21)=54123.$$

Thus,

$$\big(B_3^\circ(231,21)\big)^{RU}=B_3^\circ(231^{RU},21^{RU}),$$

as stated in the lemma. In contrast to (8.4), the two components are not interchanged here.

To analyze the maps F_k recursively, we first describe how they interact with the two insertion operations introduced above. This recursive relation is naturally coupled with a symmetry property of F_k, and the two statements will be proved simultaneously by induction.

Proposition 8.5. *The maps F_k satisfy the following two properties.*

(E_k) *Let $k\geq 3$, and suppose that*

$$F_{k-1}(\sigma,\tau)=(\alpha,\beta),$$

where $\sigma,\tau,\alpha,\beta\in S_{k-1}$. Then, for $d,h\in[k]$,

$$F_k\big(I_d^-(\sigma),I_h^+(\tau)\big)=\big(I_d^-(\beta),I_h^+(\alpha)\big).$$

(D_k) *Let $k\geq 2$, and suppose that*

$$F_k(\sigma,\tau)=(\alpha,\beta).$$

Then

$$F_k\big(\tau^{RU},\sigma^{RU}\big)=\big(\beta^{RU},\alpha^{RU}\big).$$

Proof. We prove (E_k) and (D_k) by simultaneous induction. The induction proceeds according to the following scheme:

$$D_2 \implies E_3 \implies D_3 \implies E_4 \implies D_4 \implies \cdots.$$

More precisely, we first verify (D_2). Then, for each $k \geq 3$, assuming (D_{k-1}), we first prove (E_k) and then use (E_k) together with (D_{k-1}) to prove (D_k).

It follows directly from the definition of f_4 that F_2 is the identity map on $\mathfrak{S}_2 \times \mathfrak{S}_2$. Hence (D_2) holds.

We now assume (D_{k-1}) and prove (E_k). Suppose that

$$F_{k-1}(\sigma, \tau) = (\alpha, \beta),$$

and let $d, h \in [k]$. Set

$$W = A_k\big(I_d^-(\sigma), I_h^+(\tau)\big).$$

The entry $2k$ occurs in position $k + h$ of W. Removing this entry gives a permutation $\widetilde{W} \in \widetilde{\mathfrak{A}}_{2k-1}$ such that $W = \mathrm{ins}_{k+h}(\widetilde{W})$. Therefore, by the definition of f_{2k},

$$f_{2k}(W) = \mathrm{inss}_{2(k+h-(k-1)-1)}\big(\tilde{f}_{2k-1}(\widetilde{W})\big) = \mathrm{inss}_{2h}\big(\tilde{f}_{2k-1}(\widetilde{W})\big).$$

Since $W = A_k\big(I_d^-(\sigma), I_h^+(\tau)\big)$, removing the entry $2k$ gives

$$\widetilde{W} = (\sigma_1 + 1) \cdots (\sigma_{d-1} + 1)\, 1\, (\sigma_d + 1) \cdots (\sigma_{k-1} + 1)(k + \tau_1) \cdots (k + \tau_{k-1}).$$

Applying RU to $\widetilde{W}$, the entry 1 becomes the new maximum $2k-1$ and moves from position d to position $2k - d$. Moreover,

$$\tau^{RU} = (k - \tau_{k-1}) \cdots (k - \tau_1), \text{ and } \sigma^{RU} = (k - \sigma_{k-1}) \cdots (k - \sigma_1).$$

Hence

$$\widetilde{W}^{RU} = \mathrm{ins}_{2k-d}\big(A_{k-1}(\tau^{RU}, \sigma^{RU})\big).$$

By the induction hypothesis (D_{k-1}),

$$F_{k-1}(\tau^{RU}, \sigma^{RU}) = (\beta^{RU}, \alpha^{RU}),$$

and by the definition of F_{k-1}, we have

$$f_{2k-2}\big(A_{k-1}(\tau^{RU}, \sigma^{RU})\big) = B_{k-1}(\beta^{RU}, \alpha^{RU}).$$

Set $s = k + 1 - d$. Applying the odd step in the definition of f_{2k-1}, we obtain

$$f_{2k-1}(\widetilde{W}^{RU}) = \mathrm{inss}_{2(2k-d-(k-1))-1}\big(B_{k-1}(\beta^{RU}, \alpha^{RU})\big) = \mathrm{inss}_{2s-1}\big(B_{k-1}(\beta^{RU}, \alpha^{RU})\big),$$

which equals

$$B_k^\circ\big(I_s^+(\beta^{RU}), \alpha^{RU}\big)$$

by Lemma 8.3.

By Lemma 8.2, we have $I_s^+(\beta^{RU}) = \big(I_d^-(\beta)\big)^{RU}$. Therefore,

$$f_{2k-1}(\widetilde{W}^{RU}) = B_k^\circ\big((I_d^-(\beta))^{RU}, \alpha^{RU}\big).$$

By the definition of $\tilde{f}_{2k-1}$ and Lemma 8.4, we have

$$\tilde{f}_{2k-1}(\widetilde{W}) = \big(f_{2k-1}(\widetilde{W}^{RU})\big)^{RU} = \big[B_k^\circ\big((I_d^-(\beta))^{RU}, \alpha^{RU}\big)\big]^{RU} = B_k^\circ\big(I_d^-(\beta), \alpha\big).$$

Consequently, by Lemma 8.3, we have

$$f_{2k}(W) = \mathrm{inss}_{2h}\left(B_k^\circ\big(I_d^-(\beta), \alpha\big)\right) = B_k\big(I_d^-(\beta), I_h^+(\alpha)\big).$$

Therefore, by the definition of F_k, we conclude that

$$F_k\big(I_d^-(\sigma), I_h^+(\tau)\big) = \big(I_d^-(\beta), I_h^+(\alpha)\big),$$

which proves (E_k).

We next assume (E_k) together with (D_{k-1}) to prove (D_k). Suppose that

$$F_k(\sigma, \tau) = (\alpha, \beta).$$

Let d be the position of 1 in σ, and let h be the position of k in τ. There are unique $\sigma', \tau' \in \mathfrak{S}_{k-1}$ such that

$$\sigma = I_d^-(\sigma'), \qquad \tau = I_h^+(\tau').$$

Write

$$F_{k-1}(\sigma', \tau') = (\alpha', \beta').$$

Applying (E_k), we have

$$\alpha = I_d^-(\beta'), \qquad \beta = I_h^+(\alpha').$$

By (D_{k-1}),

$$F_{k-1}\big((\tau')^{RU}, (\sigma')^{RU}\big) = \big((\beta')^{RU}, (\alpha')^{RU}\big).$$

Lemma 8.2 give

$$\tau^{RU} = I_{k+1-h}^-\big((\tau')^{RU}\big), \qquad \sigma^{RU} = I_{k+1-d}^+\big((\sigma')^{RU}\big).$$

Applying (E_k) again, $F_k(\tau^{RU}, \sigma^{RU})$ equals

$$F_k\left(I_{k+1-h}^-\big((\tau')^{RU}\big), I_{k+1-d}^+\big((\sigma')^{RU}\big)\right) = \left(I_{k+1-h}^-\big((\alpha')^{RU}\big), I_{k+1-d}^+\big((\beta')^{RU}\big)\right).$$

Using Lemma 8.2 once more,

$$I_{k+1-h}^-\big((\alpha')^{RU}\big) = \big(I_h^+(\alpha')\big)^{RU} = \beta^{RU}, \qquad I_{k+1-d}^+\big((\beta')^{RU}\big) = \big(I_d^-(\beta')\big)^{RU} = \alpha^{RU}.$$

Therefore,

$$F_k(\tau^{RU}, \sigma^{RU}) = (\beta^{RU}, \alpha^{RU}),$$

which proves (D_k) and completes the induction. □

We next recall the rank-matrix criterion for the Bruhat order. For $w = w_1 \cdots w_n \in \mathfrak{S}_n$ and $a, b \in [n]$, define

$$r_w(a, b) = |\{\, i \in [a] \mid w_i \le b \,\}|.$$

For $a = 0$ or $b = 0$, we use the convention

$$r_w(0, b) = r_w(a, 0) = 0.$$

The rank matrix of w is

$$R_w = \big(r_w(a, b)\big)_{1 \le a, b \le n}.$$

Thus, $r_w(a, b)$ counts the number of entries among the first a positions of w whose values are at most b.

The following well-known rank-matrix criterion [1, Theorem 2.1.5] characterizes the Bruhat order.

Theorem 8.6 (Rank-matrix criterion for the Bruhat order)**.** *Let $u, w \in \mathfrak{S}_n$. Then*

$$u \le w \quad \Longleftrightarrow \quad r_u(a, b) \ge r_w(a, b)$$

for all $a, b \in [n]$.

For example, consider $u = 2143$ and $w = 3241$. Their rank matrices are

$$R_u = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{bmatrix}, \qquad R_w = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{bmatrix}.$$

Since every entry of R_u is greater than or equal to the corresponding entry of R_w, the rank-matrix criterion gives $2143 \leq 3241$.

Let $w \in \mathfrak{A}_{2k}$. Write

$$w = A_k(\sigma, \tau)$$

for some $\sigma, \tau \in \mathfrak{S}_k$. Suppose that

$$f_{2k}(w) = B_k(\alpha, \beta),$$

where $\alpha, \beta \in \mathfrak{S}_k$. Equivalently, by the definition of F_k,

$$F_k(\sigma, \tau) = (\alpha, \beta).$$

Thus, to prove

$$w \leq f_{2k}(w),$$

it is enough to prove

$$A_k(\sigma, \tau) \leq B_k(\alpha, \beta).$$

By Theorem 8.6, it therefore suffices to show that

$$r_{A_k(\sigma,\tau)}(a, b) \geq r_{B_k(\alpha,\beta)}(a, b) \tag{8.7}$$

for all $a, b \in [2k]$. We will verify (8.7) by considering the four regions determined by whether a and b are at most or greater than k.

To compare the rank matrices of $A_k(\sigma, \tau)$ and $B_k(\alpha, \beta)$, it will be useful to express the rank function of $B_k(\alpha, \beta)$ directly in terms of the rank functions of α and β. The following lemma gives this decomposition.

Lemma 8.7. *Let $\alpha, \beta \in \mathfrak{S}_k$ and*

$$u = B_k(\alpha, \beta) \in \mathfrak{B}_{2k}.$$

Then, for all $a, b \in [2k]$,

$$r_u(a, b) = r_\alpha\left(\left\lceil \frac{a}{2} \right\rceil, \left\lceil \frac{b}{2} \right\rceil\right) + r_\beta\left(\left\lfloor \frac{a}{2} \right\rfloor, \left\lfloor \frac{b}{2} \right\rfloor\right).$$

Proof. Among the first a positions of $u = B_k(\alpha, \beta)$, there are $\lceil \frac{a}{2} \rceil$ odd positions and $\lfloor \frac{a}{2} \rfloor$ even positions.

The entries in the odd positions are

$$2\alpha_1 - 1, \ldots, 2\alpha_{\lceil \frac{a}{2} \rceil} - 1.$$

For each i, we know that

$$2\alpha_i - 1 \leq b \quad \Longleftrightarrow \quad \alpha_i \leq \left\lceil \frac{b}{2} \right\rceil.$$

Hence the contribution from the odd positions is $r_\alpha\left(\lceil \frac{a}{2} \rceil, \lceil \frac{b}{2} \rceil\right)$.

Similarly, the entries in the even positions are

$$2\beta_1, \ldots, 2\beta_{\lfloor \frac{a}{2} \rfloor}.$$

For each j, we know that

$$2\beta_j \le b \quad \Longleftrightarrow \quad \beta_j \le \left\lfloor \frac{b}{2} \right\rfloor.$$

Hence the contribution from the even positions is $r_\beta\left(\lfloor \frac{a}{2} \rfloor, \lfloor \frac{b}{2} \rfloor\right)$.

Adding the two contributions gives

$$r_u(a,b) = r_\alpha\left(\left\lceil \frac{a}{2} \right\rceil, \left\lceil \frac{b}{2} \right\rceil\right) + r_\beta\left(\left\lfloor \frac{a}{2} \right\rfloor, \left\lfloor \frac{b}{2} \right\rfloor\right).$$

□

By Theorem 8.6, it remains to compare the rank functions of $A_k(\sigma,\tau)$ and $B_k(\alpha,\beta)$ at every pair $(a,b) \in [2k] \times [2k]$. We divide this comparison into four regions according to whether a and b are at most k or greater than k. Before we begin with the northwest quadrant, where $a, b \in [k]$. We need some auxiliary lemmas.

Lemma 8.8. *Let $w \in \mathfrak{S}_{k-1}$.*

(i) For $d \in [k]$ and $a, b \in [k]$,

$$r_{I_d^-(w)}(a,b) = \begin{cases} r_w(a,b-1), & d > a, \\ 1 + r_w(a-1,b-1), & d \le a. \end{cases} \tag{8.8}$$

(ii) For $h \in [k]$, $a \in [k]$, and $b \in [k-1]$,

$$r_{I_h^+(w)}(a,b) = \begin{cases} r_w(a,b), & h > a, \\ r_w(a-1,b), & h \le a. \end{cases} \tag{8.9}$$

Here we use the convention $r_w(a,0) = 0$.

Proof. For (8.8), recall that $I_d^-(w)$ is obtained by increasing every entry of w by 1 and inserting 1 in position d. If $d > a$, the inserted 1 does not occur among the first a positions, and an entry of $I_d^-(w)$ is at most b precisely when the corresponding entry of w is at most $b-1$. Hence

$$r_{I_d^-(w)}(a,b) = r_w(a,b-1).$$

If $d \le a$, the inserted 1 contributes one to the rank, while the remaining $a-1$ entries contribute $r_w(a-1,b-1)$. Thus

$$r_{I_d^-(w)}(a,b) = 1 + r_w(a-1,b-1).$$

For (8.9), $I_h^+(w)$ is obtained by inserting the new maximum k in position h. Since $b \le k-1$, the inserted entry k does not contribute to the rank. If $h > a$, the first a entries are unchanged, while if $h \le a$, only $a-1$ entries of w occur among the first a positions. Therefore,

$$r_{I_h^+(w)}(a,b) = \begin{cases} r_w(a,b), & h > a, \\ r_w(a-1,b), & h \le a. \end{cases}$$

□

Lemma 8.9. *Let $\sigma, \tau, \alpha, \beta \in \mathfrak{S}_k$, and suppose that*

$$F_k(\sigma, \tau) = (\alpha, \beta).$$

Then, for all $a, b \in [k]$ and $0 \le p \le a$,

$$r_\sigma(a,b) \ge r_\alpha\left(p, \left\lceil \frac{b}{2} \right\rceil\right) + r_\beta\left(a-p, \left\lfloor \frac{b}{2} \right\rfloor\right). \tag{R_k}$$

Proof. Before we start, we call a, b, p the first, second, third parameter of (R_k). Now, we proceed by induction on k. We use the convention $r_w(0,b) = r_w(a,0) = 0$. With this convention, the statement is also trivial when one of the parameters occurring in the induction hypothesis is zero.

For $k = 2$, by Algorithm 7.15, the map F_2 is the identity map on $\mathfrak{S}_2 \times \mathfrak{S}_2$. Thus $\alpha = \sigma$ and $\beta = \tau$. If $b = 1$, then

$$r_\alpha(p,1) + r_\beta(a-p,0) = r_\sigma(p,1) \le r_\sigma(a,1).$$

If $b = 2$, then

$$r_\alpha(p,1) + r_\beta(a-p,1) \le p + (a-p) = a = r_\sigma(a,2).$$

Hence (R_2) holds.

Now let $k \ge 3$ and assume (R_{k-1}). Suppose that

$$F_k(\sigma, \tau) = (\alpha, \beta).$$

Let d be the position of 1 in σ, and let h be the position of k in τ. Then there exist $\sigma', \tau' \in \mathfrak{S}_{k-1}$ such that

$$\sigma = I_d^-(\sigma'), \qquad \tau = I_h^+(\tau').$$

Write

$$F_{k-1}(\sigma', \tau') = (\alpha', \beta').$$

By (E_k) in Proposition 8.5,

$$\alpha = I_d^-(\beta'), \qquad \beta = I_h^+(\alpha').$$

Fix $a, b \in [k]$ and $0 \le p \le a$. Define

$$\delta = \mathbf{1}_{d \le a}, \qquad \eta = \mathbf{1}_{d \le p}, \qquad \varepsilon = \mathbf{1}_{h \le a-p}.$$

Since $p \le a$, we always have $\eta \le \delta$.

By (8.8) and (8.9), we have

$$r_\sigma(a,b) = \delta + r_{\sigma'}(a-\delta, b-1), \tag{8.10}$$

$$r_\alpha\left(p, \left\lceil \frac{b}{2} \right\rceil\right) = \eta + r_{\beta'}\left(p-\eta, \left\lceil \frac{b}{2} \right\rceil - 1\right), \tag{8.11}$$

$$r_\beta\left(a-p, \left\lfloor \frac{b}{2} \right\rfloor\right) = r_{\alpha'}\left(a-p-\varepsilon, \left\lfloor \frac{b}{2} \right\rfloor\right). \tag{8.12}$$

Now, set $q = a - p - \varepsilon$.

We first consider the case $(\delta, p, \varepsilon) \ne (1,0,0)$. Then, we have $0 \le q \le a - \delta$. Indeed, $q \ge 0$ by the definition of ε, while $q \le a - \delta$ is equivalent to $p + \varepsilon \ge \delta$, which fails only when $(\delta, p, \varepsilon) = (1,0,0)$.

Note that $(a-\delta)-q=(a-\delta)-(a-p-\varepsilon)=p+\varepsilon-\delta$. By the definition of δ, we have

$$a-\delta\leq\begin{cases}k-\delta=k-1, & \text{when } a=k,\\ k-1-\delta\leq k-1, & \text{when } a<k.\end{cases}$$

We also note that $b-1\leq k-1$. Applying (R_{k-1}) with parameters $a-\delta, b-1, q$ and using

$$\left\lceil\frac{b-1}{2}\right\rceil=\left\lfloor\frac{b}{2}\right\rfloor,\qquad\left\lfloor\frac{b-1}{2}\right\rfloor=\left\lceil\frac{b}{2}\right\rceil-1,$$

we obtain

$$r_{\sigma'}(a-\delta,b-1)\geq r_{\alpha'}\left(q,\left\lfloor\frac{b}{2}\right\rfloor\right)+r_{\beta'}\left(p+\varepsilon-\delta,\left\lceil\frac{b}{2}\right\rceil-1\right).$$

Comparing this inequality with (8.10), we have

$$r_{\sigma}(a,b)\geq r_{\alpha'}\left(q,\left\lfloor\frac{b}{2}\right\rfloor\right)+\delta+r_{\beta'}\left(p+\varepsilon-\delta,\left\lceil\frac{b}{2}\right\rceil-1\right).\tag{8.13}$$

Next, we claim that

$$\delta+r_{\beta'}\left(p+\varepsilon-\delta,\left\lceil\frac{b}{2}\right\rceil-1\right)\geq\eta+r_{\beta'}\left(p-\eta,\left\lceil\frac{b}{2}\right\rceil-1\right).\tag{8.14}$$

To see this, we first note that $\delta\geq\eta$ by above definition. Then, it remains to compare the other two terms. If $p-\eta\leq p+\varepsilon-\delta$, (8.14) follows from the monotonicity of the rank function in its first argument. Otherwise, we have $p-\eta>p+\varepsilon-\delta$, i.e., $\eta<\delta-\varepsilon$. Hence

$$\delta=1,\qquad\eta=\varepsilon=0,$$

and the claim (8.14) follows from $r_{\beta'}(x+1,y)\leq r_{\beta'}(x,y)+1$ for all (possible) x and y.

Comparing the right-hand-side of (8.14) with (8.11) and by (8.12), we have

$$r_{\sigma}(a,b)\geq r_{\alpha'}\left(q,\left\lfloor\frac{b}{2}\right\rfloor\right)+\eta+r_{\beta'}\left(p-\eta,\left\lceil\frac{b}{2}\right\rceil-1\right)=r_{\beta}\left(a-p,\left\lfloor\frac{b}{2}\right\rfloor\right)+r_{\alpha}\left(p,\left\lceil\frac{b}{2}\right\rceil\right).$$

It remains to consider the boundary case $(\delta,p,\varepsilon)=(1,0,0)$. Since $p=0$, by the definition of η, we have $\eta=0$. Since $\varepsilon=0$ and $p=0$, we have $h>a$. Therefore, (8.10) and (8.12) become

$$r_{\sigma}(a,b)=1+r_{\sigma'}(a-1,b-1)$$

and

$$r_{\beta}\left(a,\left\lfloor\frac{b}{2}\right\rfloor\right)=r_{\alpha'}\left(a,\left\lfloor\frac{b}{2}\right\rfloor\right).$$

Indeed, $r_{\alpha'}\left(a,\left\lfloor\frac{b}{2}\right\rfloor\right)$ is well-defined since $h>a$ implies $a\leq k-1$. Applying (R_{k-1}) such that the third parameter equals $a-1$, we obtain

$$r_{\sigma'}(a-1,b-1)\geq r_{\alpha'}\left(a-1,\left\lfloor\frac{b}{2}\right\rfloor\right)+r_{\beta'}\left(0,\left\lceil\frac{b}{2}\right\rceil-1\right)=r_{\alpha'}\left(a-1,\left\lfloor\frac{b}{2}\right\rfloor\right).$$

Since increasing the first argument of a rank function by one increases its value by at most one, we have

$$1+r_{\alpha'}\left(a-1,\left\lfloor\frac{b}{2}\right\rfloor\right)\geq r_{\alpha'}\left(a,\left\lfloor\frac{b}{2}\right\rfloor\right).$$

Hence, by (8.10)

$$r_\sigma(a,b) \geq r_{\alpha'}\left(a, \left\lfloor \frac{b}{2} \right\rfloor\right).$$

Since $\beta = I_h^+(\alpha')$ and $h > a$, inserting the maximum entry at position h does not change the rank function. We have

$$r_{\alpha'}\left(a, \left\lfloor \frac{b}{2} \right\rfloor\right) = r_{I_h^+(\alpha')}\left(a, \left\lfloor \frac{b}{2} \right\rfloor\right) = r_\beta\left(a, \left\lfloor \frac{b}{2} \right\rfloor\right).$$

Hence, we have

$$r_\sigma(a,b) \geq r_\beta\left(a, \left\lfloor \frac{b}{2} \right\rfloor\right)$$

Since $r_\alpha\left(0, \lceil \frac{b}{2} \rceil\right) = 0$, this is precisely (R_k) when $p = 0$.

Thus (R_k) holds, completing the induction. □

The estimate (R_k) in Lemma 8.9 is stronger than what is needed for the northwest quadrant. Indeed, the latter is obtained by choosing $p = \lceil \frac{a}{2} \rceil$ in (R_k). This immediately gives the following corollary.

Corollary 8.10 (Northwest quadrant)**.** *Let $\sigma, \tau, \alpha, \beta \in \mathfrak{S}_k$, and suppose that*

$$F_k(\sigma, \tau) = (\alpha, \beta).$$

Then, for all $a, b \in [k]$,

$$r_{A_k(\sigma,\tau)}(a,b) \geq r_{B_k(\alpha,\beta)}(a,b).$$

Proof. Since $a, b \in [k]$, by the definition of A_k,

$$r_{A_k(\sigma,\tau)}(a,b) = r_\sigma(a,b).$$

Applying (R_k) in Lemma 8.9 with $p = \lceil \frac{a}{2} \rceil$, we obtain

$$r_\sigma(a,b) \geq r_\alpha\left(\left\lceil \frac{a}{2} \right\rceil, \left\lceil \frac{b}{2} \right\rceil\right) + r_\beta\left(\left\lfloor \frac{a}{2} \right\rfloor, \left\lfloor \frac{b}{2} \right\rfloor\right),$$

where we have used

$$a - \left\lceil \frac{a}{2} \right\rceil = \left\lfloor \frac{a}{2} \right\rfloor.$$

By Lemma 8.7, the right-hand side is precisely

$$r_{B_k(\alpha,\beta)}(a,b).$$

Hence

$$r_{A_k(\sigma,\tau)}(a,b) \geq r_{B_k(\alpha,\beta)}(a,b),$$

as desired. □

The northwest quadrant has now been settled. The two mixed quadrants are considerably simpler: in these regions, the desired rank inequalities follow directly from the block structure of $A_k(\sigma, \tau)$. We record these two cases together in the following lemma.

Lemma 8.11 (Mixed quadrants)**.** *Let $\sigma, \tau, \alpha, \beta \in \mathfrak{S}_k$, and suppose that*

$$F_k(\sigma, \tau) = (\alpha, \beta).$$

Then

$$r_{A_k(\sigma,\tau)}(a, b) \geq r_{B_k(\alpha,\beta)}(a, b)$$

whenever either $a \leq k < b$ or $b \leq k < a$.

Proof. First suppose that $a \leq k < b$. Since the first a entries of $A_k(\sigma, \tau)$ all belong to $[k]$, they are all at most b. Therefore, $r_{A_k(\sigma,\tau)}(a, b) = a$.

On the other hand, for any permutation w we have $r_w(a, b) \leq a$. Hence

$$r_{A_k(\sigma,\tau)}(a, b) = a \geq r_{B_k(\alpha,\beta)}(a, b).$$

Now suppose that $b \leq k < a$. Among the first k entries of $A_k(\sigma, \tau)$, each element of $[k]$ occurs exactly once, while all entries after position k are greater than k. Thus exactly b of the first a entries are at most b, and hence $r_{A_k(\sigma,\tau)}(a, b) = b$. Since $r_w(a, b) \leq b$ for every permutation w, we obtain

$$r_{A_k(\sigma,\tau)}(a, b) = b \geq r_{B_k(\alpha,\beta)}(a, b).$$

□

The northwest and the two mixed quadrants have now been settled. To handle the remaining southeast quadrant, we will relate the rank function in the southeast region to the rank function of the *RU*-transform in the northwest region. We first record the following general identity.

Lemma 8.12. *Let $w \in \mathfrak{S}_{2k}$. Then, for all $a, b \in [2k]$,*

$$r_w(a, b) = a + b - 2k + r_{w^{RU}}(2k - a, 2k - b). \tag{8.15}$$

Here we use the convention $r_w(0, b) = r_w(a, 0) = 0$.

Proof. Among the $2k$ entries of w, exactly $2k - b$ are greater than b. Let N denote the number of such entries occurring among the first a positions. Then

$$r_w(a, b) = a - N.$$

Write $w^{RU} = v_1 \cdots v_{2k}$. By the definition of *RU*, $v_j = 2k + 1 - w_{2k+1-j}$ for every $j \in [2k]$.

Now consider an entry w_i of w with $i > a$ and $w_i > b$. Under *RU*, this entry corresponds to the entry

$$v_{2k+1-i} = 2k + 1 - w_i$$

of w^{RU}. Since $i > a$, we have

$$2k + 1 - i \leq 2k - a,$$

and since $w_i > b$, we have

$$2k + 1 - w_i \leq 2k - b.$$

Thus, each entry of w greater than b and occurring after position a corresponds to an entry of w^{RU} among its first $2k - a$ positions whose value is at most $2k - b$.

Conversely, suppose that v_j is among the first $2k - a$ positions of w^{RU} with $v_j \leq 2k - b$. The corresponding entry of w is

$$w_{2k+1-j} = 2k + 1 - v_j.$$

Since $j \leq 2k - a$, we have

$$2k + 1 - j > a,$$

and since $v_j \leq 2k - b$, we have

$$2k + 1 - v_j > b.$$

Therefore, this correspondence is a bijection between the entries of w greater than b occurring after position a and the entries of w^{RU} among its first $2k - a$ positions whose values are at most $2k - b$.

Hence the number of entries of w greater than b occurring after position a is

$$r_{w^{RU}}(2k - a, 2k - b).$$

Since the total number of entries of w greater than b is $2k - b$, we obtain

$$N = 2k - b - r_{w^{RU}}(2k - a, 2k - b).$$

Substituting this into $r_w(a, b) = a - N$ gives

$$r_w(a, b) = a + b - 2k + r_{w^{RU}}(2k - a, 2k - b).$$

□

We now return to the southeast quadrant. Lemma 8.12 allows us to transfer the desired inequality to the northwest quadrant, while the symmetry (D_k) ensures that the corresponding RU-transformed permutations are still related by F_k. We can therefore complete the final quadrant comparison.

Lemma 8.13 (Southeast quadrant)**.** *Let $\sigma, \tau, \alpha, \beta \in \mathfrak{S}_k$, and suppose that*

$$F_k(\sigma, \tau) = (\alpha, \beta).$$

Then, for all $a, b \in [k + 1, 2k]$, we have

$$r_{A_k(\sigma,\tau)}(a, b) \geq r_{B_k(\alpha,\beta)}(a, b).$$

Proof. Fix $a, b \in [k + 1, 2k]$, and set

$$\overline{a} = 2k - a, \qquad \overline{b} = 2k - b.$$

Then we have $0 \leq \overline{a}, \overline{b} \leq k - 1$.

By Lemma 8.1, we have

$$A_k(\sigma, \tau)^{RU} = A_k(\tau^{RU}, \sigma^{RU}), \text{ and } B_k(\alpha, \beta)^{RU} = B_k(\beta^{RU}, \alpha^{RU}).$$

Moreover, by (D_k) in Proposition 8.5,

$$F_k(\tau^{RU}, \sigma^{RU}) = (\beta^{RU}, \alpha^{RU}).$$

If $\overline{a}, \overline{b} \geq 1$, Corollary 8.10 applied to

$$F_k(\tau^{RU}, \sigma^{RU}) = (\beta^{RU}, \alpha^{RU})$$

gives

$$r_{A_k(\tau^{RU},\sigma^{RU})}(\overline{a}, \overline{b}) \geq r_{B_k(\beta^{RU},\alpha^{RU})}(\overline{a}, \overline{b}).$$

If either $\overline{a} = 0$ or $\overline{b} = 0$, both sides are zero. Thus, in all cases,

$$r_{A_k(\sigma,\tau)^{RU}}(\overline{a}, \overline{b}) \geq r_{B_k(\alpha,\beta)^{RU}}(\overline{a}, \overline{b}). \tag{8.16}$$

Applying (8.15) to $A_k(\sigma,\tau)$, we obtain

$$r_{A_k(\sigma,\tau)}(a,b) = a+b-2k+r_{A_k(\sigma,\tau)^{RU}}(\overline{a},\overline{b}).$$

Similarly,

$$r_{B_k(\alpha,\beta)}(a,b) = a+b-2k+r_{B_k(\alpha,\beta)^{RU}}(\overline{a},\overline{b}).$$

Therefore, by (8.16), we have

$$\begin{aligned} r_{A_k(\sigma,\tau)}(a,b) &= a+b-2k+r_{A_k(\sigma,\tau)^{RU}}(\overline{a},\overline{b}) \\ &\geq a+b-2k+r_{B_k(\alpha,\beta)^{RU}}(\overline{a},\overline{b}) \\ &= r_{B_k(\alpha,\beta)}(a,b). \end{aligned}$$

This proves the southeast quadrant. □

Combining the northwest, mixed, and southeast quadrant comparisons, we immediately obtain the following corollary.

Corollary 8.14. *Let $\sigma,\tau,\alpha,\beta \in \mathfrak{S}_k$, and suppose that*

$$F_k(\sigma,\tau) = (\alpha,\beta).$$

Then, for all $a,b \in [2k]$, we have

$$r_{A_k(\sigma,\tau)}(a,b) \geq r_{B_k(\alpha,\beta)}(a,b).$$

The preceding rank comparisons establish the required inequality at every entry of the rank matrix. We can therefore apply the rank-matrix criterion to obtain the even case of the main result.

Theorem 8.15. *Let $n = 2k$ ($k \geq 2$). Then, for every $w \in \mathfrak{A}_{2k}$,*

$$w \leq f_{2k}(w)$$

under the Bruhat order.

Proof. This follows immediately from Corollary 8.14 and Theorem 8.6. □

The odd case follows directly from the even case. More precisely, the result for $n = 2k$ implies the corresponding result for $n = 2k-1$.

Corollary 8.16. *Let $n = 2k-1$ ($k \geq 3$). Then, for every $w \in \mathfrak{A}_{2k-1}$,*

$$w \leq f_{2k-1}(w)$$

under the Bruhat order.

Proof. Let $w \in \mathfrak{A}_{2k-1}$. Since $w^{RU} \in \widetilde{\mathfrak{A}}_{2k-1}$, we have

$$\mathrm{ins}_{2k}(w^{RU}) \in \mathfrak{A}_{2k}.$$

By the definition of $\widetilde{f}_{2k-1}$,

$$\widetilde{f}_{2k-1}(w^{RU}) = \big[f_{2k-1}\big((w^{RU})^{RU}\big)\big]^{RU} = f_{2k-1}(w)^{RU}.$$

Since inss_{2k} inserts $2k$ at the last position, no swapping occurs, and hence $\mathrm{inss}_{2k} = \mathrm{ins}_{2k}$. Therefore, by the definition of f_{2k},

$$f_{2k}\big(\mathrm{ins}_{2k}(w^{RU})\big) = \mathrm{ins}_{2k}\big(f_{2k-1}(w)^{RU}\big).$$

By Theorem 8.15,

$$\mathrm{ins}_{2k}(w^{RU}) \leq \mathrm{ins}_{2k}\big(f_{2k-1}(w)^{RU}\big).$$

Both permutations have the common last entry $2k$. By the tableau criterion (Theorem 7.23), deleting this common last entry gives

$$w^{RU} \leq f_{2k-1}(w)^{RU}.$$

Since RU preserves the Bruhat order (Proposition 7.9) and is an involution, we conclude that

$$w \leq f_{2k-1}(w).$$

□

Combining the even and odd cases, we obtain the main result of this section.

Theorem 8.17. *Let f_n be the map defined in Algorithm 7.15. For every $n \geq 4$ and every $w \in \mathfrak{A}_n$, we have*

$$w \leq f_n(w)$$

under the Bruhat order.

Proof. For even n, the conclusion follows from Theorem 8.15. For odd $n \geq 5$, it follows from Corollary 8.16. □

By Theorem 7.1 and Theorem 8.17, we immediately get the balanced case of (1.9).

Theorem 8.18. *Let A be an $n \times n$ TNN matrix. Then the inequality*

$$\mathrm{per}(A_{[1,h],[1,h]})\,\mathrm{per}(A_{[h+1,n],[h+1,n]}) \;\geq\; \mathrm{per}(A_{[1,n]\cap 2\mathbb{Z},[1,n]\cap 2\mathbb{Z}})\,\mathrm{per}(A_{[1,n]\smallsetminus 2\mathbb{Z},[1,n]\smallsetminus 2\mathbb{Z}})$$

holds when $h = \lfloor \frac{n}{2} \rfloor$ or $\lceil \frac{n}{2} \rceil$.

9. Open Problems

Results in Sections 1-8 suggest several problems for further investigation.

It is reasonable to assume that a method similar to that used in the proof of Theorem 5.1, i.e., a merge function more general than that defined in Section 4, could be used to prove the following.

Conjecture 9.1. *Let A be the antiadjacency matrix of an n-element unit interval order. Define sets*

$$I = [1,n]_0^3, \quad J = [1,n]_1^3, \quad K = [1,n]_2^3.$$

Then, for all $1 \leq h_1 < h_2 \leq n$, we have

$$\mathrm{per}(A_{[1,h_1],[1,h_1]})\,\mathrm{per}(A_{[h_1,h_2],[h_1,h_2]})\,\mathrm{per}(A_{[h_2,n],[h_2,n]}) \geq \mathrm{per}(A_{I,I})\,\mathrm{per}(A_{J,J})\,\mathrm{per}(A_{K,K}).$$

Although Theorem 5.2 does not hold for all TNN matrices (Section 5), it would be interesting to characterize the TNN matrices that do satisfy Theorem 5.2.

Problem 9.2. *What kind of TNN matrices satisfy the inequality in Theorem 5.2?*

Next, we conjecture that (5.1) holds for all TNN matrices.

Conjecture 9.3. *Let A be a TNN matrix. Fix $h \in [n]$ and define sets*

$$I = [h], \qquad J = [n] \cap 2\mathbb{Z}.$$

Then we have

$$\operatorname{per}(A_{I,I}) \operatorname{per}(A_{\bar{I},\bar{I}}) \geq \operatorname{per}(A_{J,J}) \operatorname{per}(A_{\bar{J},\bar{J}}).$$

Finally, we would be interested to see which sets satisfying determinantal and permanental inequalities for TNN matrices correspond bijectively, with inequality directions reversed.

Problem 9.4. *Decide which sets $L, K \subseteq [n]$ satisfy*

$$\operatorname{per}(A_{L,L}) \operatorname{per}(A_{\overline{L},\overline{L}}) \geq \operatorname{per}(A_{K,K}) \operatorname{per}(A_{\overline{K},\overline{K}})$$

and

$$\det(A_{L,L}) \det(A_{\overline{L},\overline{L}}) \leq \det(A_{K,K}) \det(A_{\overline{K},\overline{K}})$$

for all TNN matrices.

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10. Appendix

Here is a MATLAB code to generate examples from Theorem 8.17. `https://github.com/SihongPan/General-TNN-Matrices-Per-Top-Bottom--Per-Odd-Even-Test/blob/main/General_TNN_Matrices_Per_Ineq.m`

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